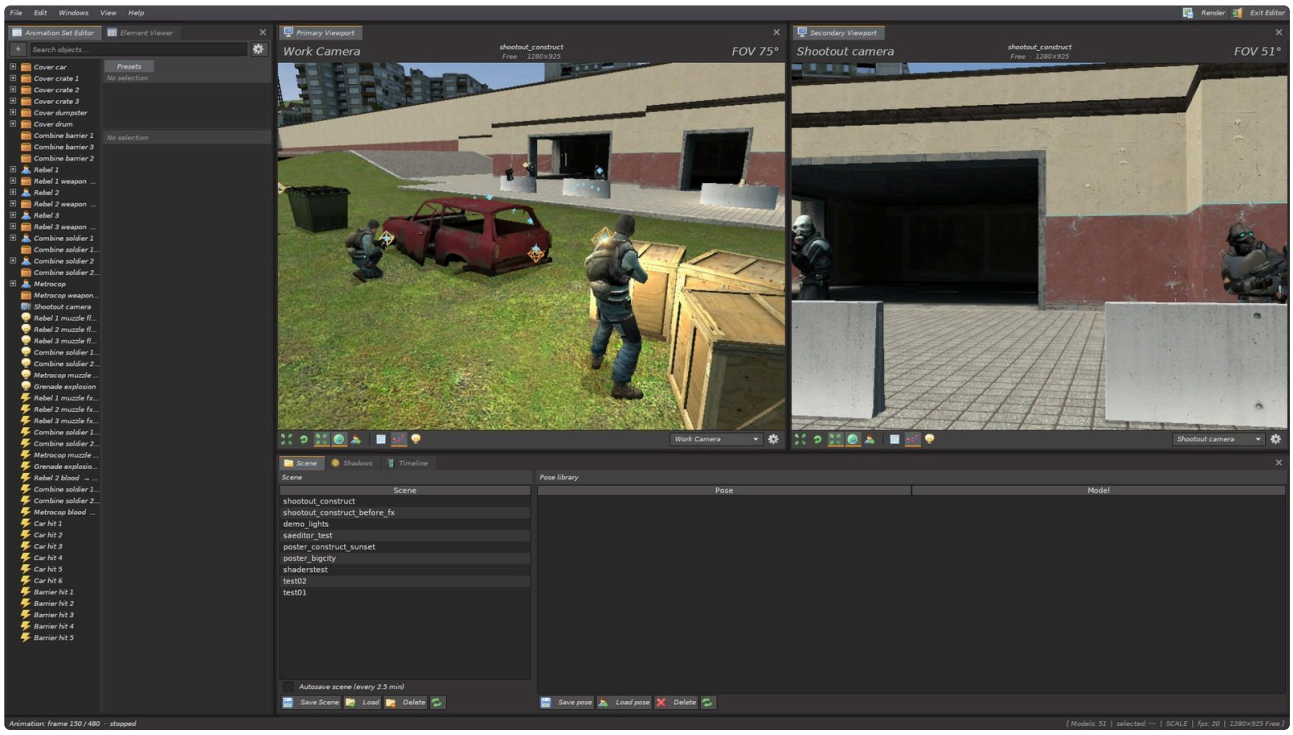
SPeditor

The complete guide



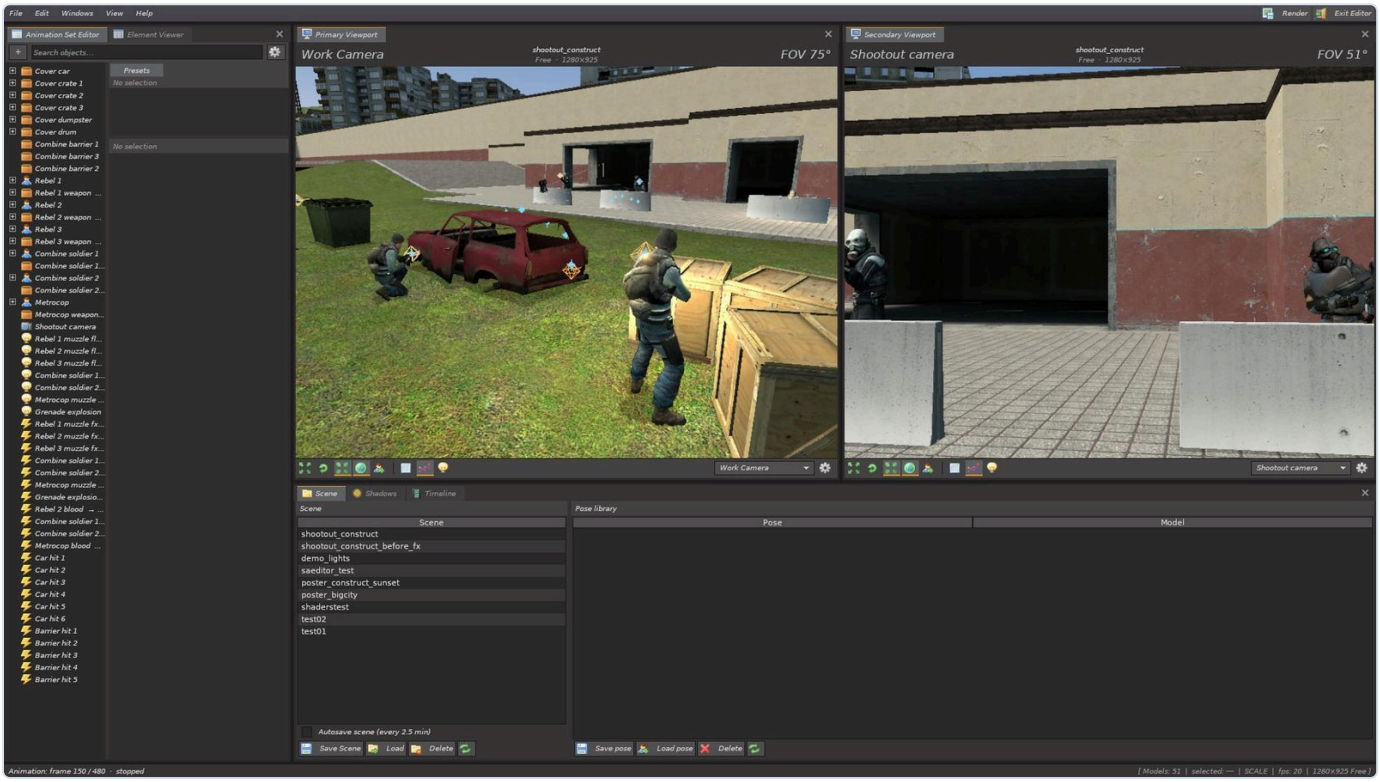
Poster, scene and animation studio for Garry's Mod · version a0.4 · October 2026
steamcommunity.com/sharedfiles/filedetails/?id=3817852435

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2. Installing and opening
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SPeditor Wiki

SPeditor turns Garry's Mod into a small film studio: posters, still scenes and short animations, built on any map with the working habits of Source Filmmaker. This wiki explains every window, button and hotkey.



The editor. Animation Set Editor on the left, the Primary and Secondary viewports in the middle and on the right, the Scene / Shadows / Timeline panel at the bottom. The scene is the bundled example `shootout_construct`.

Where to start

Installing and opening

Subscribe, open the editor, server access.

Your first poster

From an empty map to a PNG in five steps.

The workspace

Windows, tabs, docking and layouts.

Posing characters

Gizmo, bones, IK, faces, pose library.

Lights

Spot and point lights, 19 light kinds, soft shadows.

Rendering

Up to 3840 px, depth of field, camera effects.

Animation (SAeditor)

Keys, timeline, imported animations, video.

Problems and answers

The questions players ask most.

What SPeditor can do

- **An SFM-like workspace:** dockable windows, tabs you can drag out, saved layouts, eight interface themes.
- **Two viewports** with a frame gate, composition guides and fixed exposure; a free work camera and scene cameras you can look through.
- **Posing:** a gizmo for objects and for every single bone (fingers too), inverse kinematics, face flexes, eye target, skins and bodygroups, a pose library.
- **Objects:** colour and opacity, texture replacement, stretching, attaching a weapon to a hand, wearing clothing that bends with the body.
- **Lights:** point and spot lights, soft edges and penumbra, glow and a visible beam, light masks, 19 ready-made kinds, a three-point rig.
- **Effects:** explosions, fire, smoke, blood and sparks from the game and from Workshop addons, frozen at any moment.
- **Render** to PNG or JPEG up to 3840 px with smoothing, depth of field, soft shadows, camera effects and Workshop shaders.
- **SAeditor:** keyframe animation of everything in the scene, imported model animations, sounds and voices, video export.
- Scenes, poses and layouts are saved; undo and redo for every edit; English and Russian interface; a mod system for authors.

Made with SPeditor






This wiki describes SPeditor **a0.4** (alpha); the changes of this version are marked NEW. The screenshots show the English interface; switch the language in the top right corner of this page.

Installing and opening

SPeditor is a normal Workshop addon. It works in singleplayer and on servers, on any map.

Installing

1. Open the SPeditor page in the Steam Workshop and press **Subscribe**.
2. Start Garry's Mod (or restart it if it was running) and load any map. `gm_construct` and `gm_flatgrass` are good for a start.
3. Optional: subscribe to GShader Library if you want Workshop shaders in your renders (see Shaders).

Opening the editor

There are three ways. All of them toggle: the same action closes the editor again.

C menu

Hold `C` and click the **SPeditor** icon on the desktop of the context menu.

Q menu

Hold `Q`, open the **SPeditor** tab on the right, the **SPeditor** item, then **Toggle editor**.

Console

Type `speditor_toggle`. Bind it to a key to open the editor in one press: `bind f6 speditor_toggle`.



Q menu → SPeditor. The **Language** and **Theme** items next to it choose the interface language and look.

To leave the editor press ▸ **Exit Editor** in the top right corner, use ▸ **File → Exit Editor**, or run `speditor_toggle` again. Your scene stays on the map.

Interface language

Q menu → SPeditor → **Language**, or the console: `speditor_language en` / `speditor_language ru`. The language is applied the next time the editor is opened.

On a server

By default only admins may use the editor on a server. The server owner changes this with `speditor_access`:

VALUE	WHO MAY OPEN THE EDITOR
`admin`	Admins and superadmins (default)
`superadmin`	Superadmins only
`all`	Every player

Each player has a limit of models, lights, cameras and effects. A player without access sees the message “You do not have access to SPeditor on this server”.

Where your files are

Everything SPeditor saves is in `garrysmod/data/speditor/`:

FOLDER	CONTENTS
`renders/`	Pictures, frame sequences and videos you rendered
`scenes/<SteamID>/`	Saved scenes and autosaves (on a server they are kept on the server)
`poses/`	The pose library
`layouts/`	Window layouts; `_last.json` is the layout at the moment you closed the editor
`sounds/`	Your own WAV, MP3 and OGG files for SAeditor
`windows.json`	Window properties such as “Always on top”

> **!** Garry's Mod lets addons write only a few file types. Videos are therefore saved as `name.avi.dat`: remove `.dat` from the name and it is a regular AVI file.

Your first poster in five steps

A short walk through the whole workflow. Every step links to the page that explains it in detail.

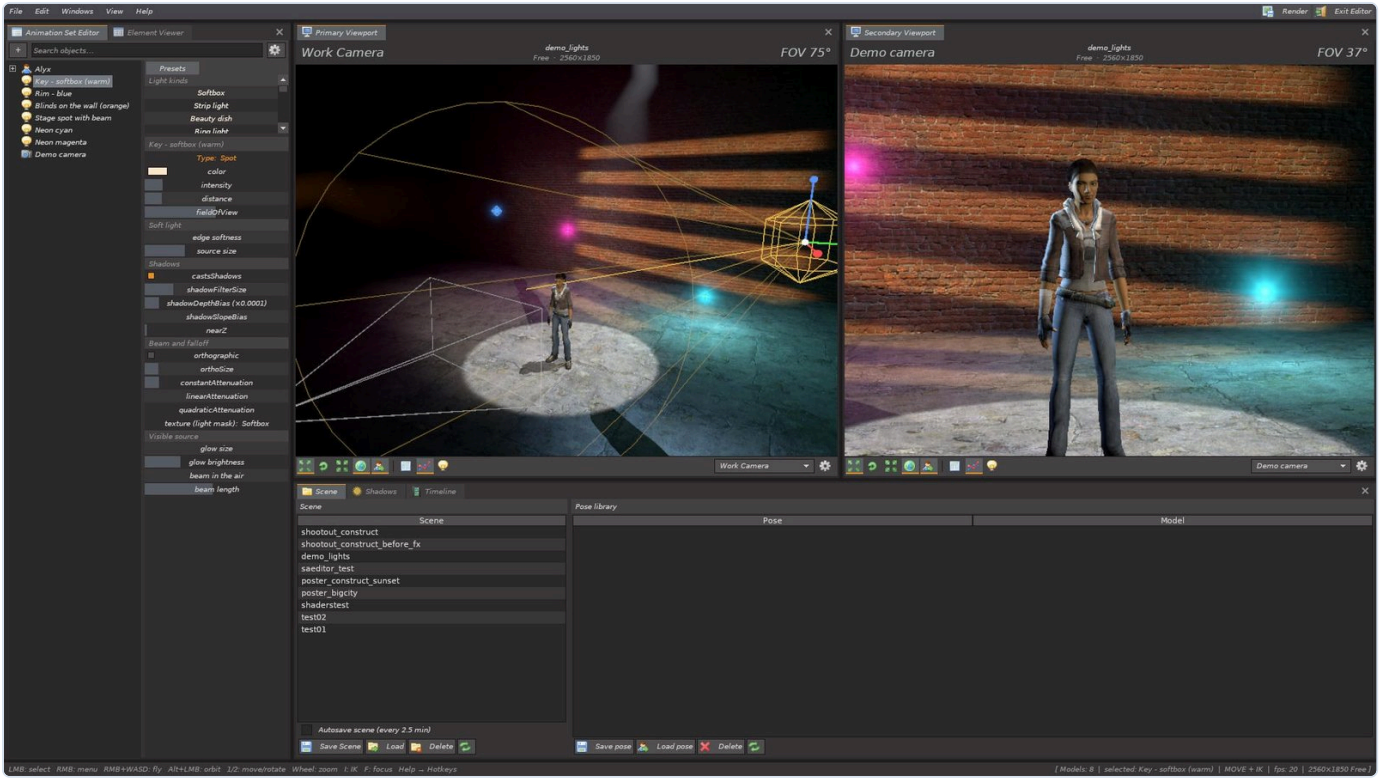
1. **Add a character.** Press **+** at the top of the Animation Set Editor and choose ▸ **Model...**. Find a model in the Model Browser (for example `alyx`) and double-click it. It appears in front of the work camera.

2. **Pose it.** Click the model in the viewport. Press `1` to move, `2` to rotate. Click a bone marker (a small cube on a joint) to pose a single bone; press `I` for inverse kinematics and drag a hand. See Posing.

3. **Light it.** **+** → ▸ **Soft light** → ▸ **Softbox** for the key light, then a ▸ **Rim light** behind the character. Or select the model and use ▸ **Create 3-point light rig**. See Lights.

4. **Frame it.** Fly the work camera to a good view (hold `RMB` and use `W` `A` `S` `D`), then **+** → ▸ **Spawn Camera**. The camera is created exactly where you look. Choose a frame aspect such as ▸ **2:3 (poster)** in the ▸ **View** → **Frame aspect** menu. See Cameras.

5. **Render.** Press ▸ **Render** in the top right corner, pick your camera, the size (up to 3840 px) and the format, and press ▸ **Render**. The file appears in `garrysmod/data/speditor/renders/`. See Rendering.



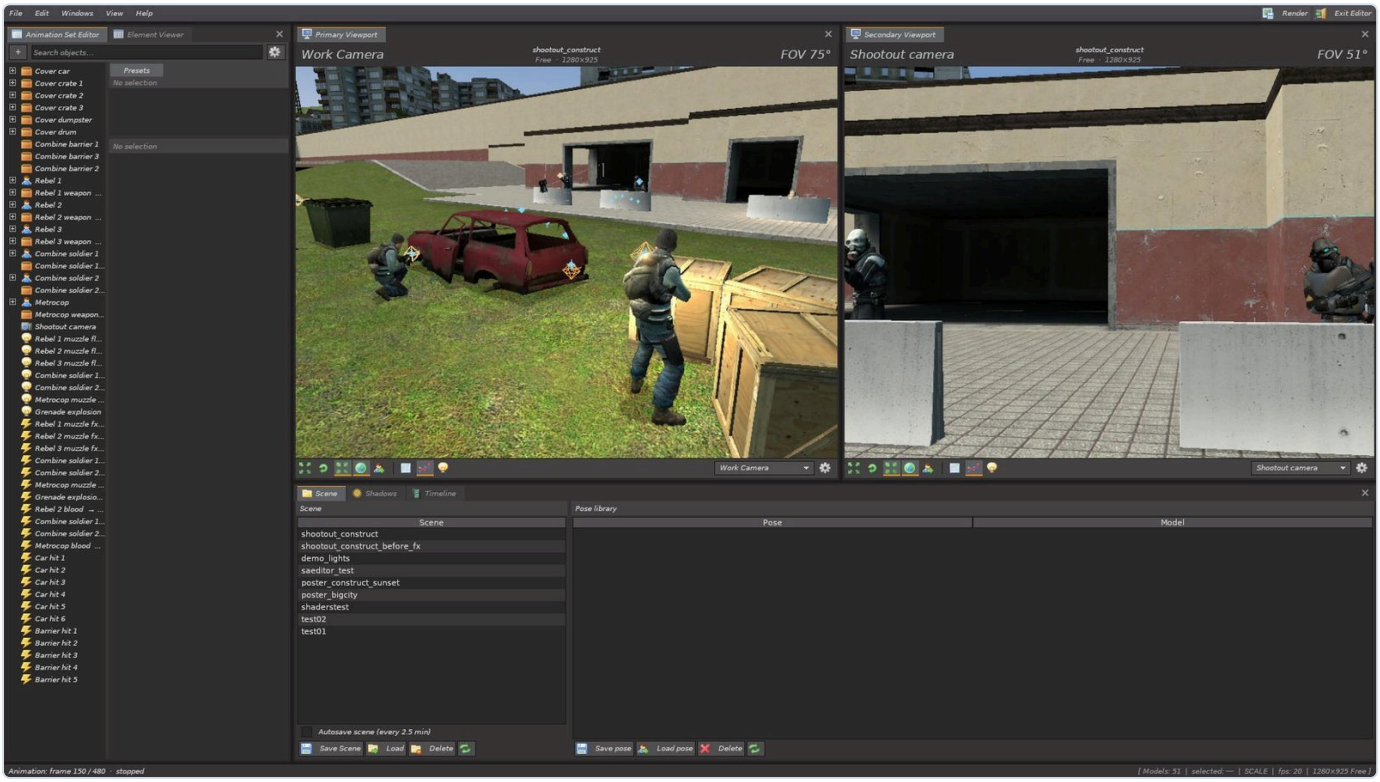
The result of steps 1–4. Left: the work camera shows the light cones and the selected softbox with its gizmo. Right: the Secondary Viewport looks through the scene camera — this is what the render will show.

Save often: ▸ **File → Save Scene...**. A scene keeps the objects, the poses, the lights, the camera effects, the frame settings and the animation. Switch on ▸ **Autosave scene** in the same menu for a backup every 2.5 minutes.

Want to see a finished scene first? Open the **Scene** tab at the bottom and load `shootout_construct` (on the map gm_construct). It is an animated shootout with effects and sounds that comes with the addon.

The workspace

The editor covers the screen with dockable windows, like Source Filmmaker. Each window holds one or more tabs; tabs can be moved between windows or torn off into a window of their own.



The default layout (“SFM default”).

Parts of the screen

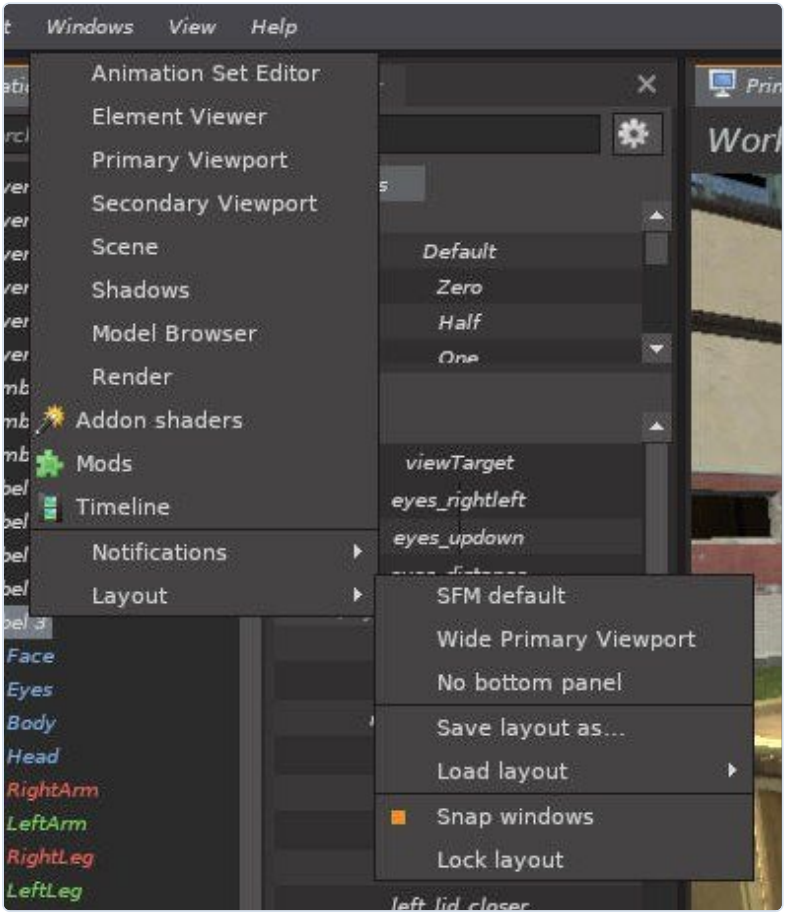
PART	WHAT IT IS FOR
Menu bar	**File, Edit, Windows, View, Help** on the left; **Render** and **Exit Editor** on the right.
Animation Set Editor	The object tree and the controls of the selected object: bones, face, eyes, light and camera settings. See Objects.
Element Viewer	Every property of the selected object as a list of numbers you can type in.
Primary Viewport	The main 3D view where you select and edit. Shows the work camera or a scene camera.
Secondary Viewport	A second view, usually through the scene camera, so you see the final frame while you work.

Scene · Shadows · Timeline	The bottom window: scene and pose library, shadow settings, and the SAeditor timeline.
Status bar	Hints on the left; on the right the number of models, the selection, the tool, FPS and the render size.

Windows and tabs

- **Move a tab:** drag it by its tab bar and drop it on the edge of another window (it docks beside it) or on another window's tab bar (it becomes a tab there). Hold `Ctrl` while dropping to keep it floating.
- **Maximize:** double-click the tab bar. Double-click again to restore.
- **Resize:** drag the gap between two docked windows; neighbours follow.
- **Window properties:** right-click the tab bar → ▸ **Window properties** → ▸ **Always on top** keeps the window above the others (the Model Browser is on top by default).
- **Reopen a closed window:** the ▸ **Windows** menu lists all of them.

Layouts



▸ **Windows → Layout** has three presets: **SFM default**, **Wide Primary Viewport** (one large viewport) and **No bottom panel**. ▸ **Save layout as...** keeps your own arrangement under a name; ▸ **Load layout** brings it back.

Snap windows makes floating windows stick to edges. **Lock layout** stops windows from being moved by accident.

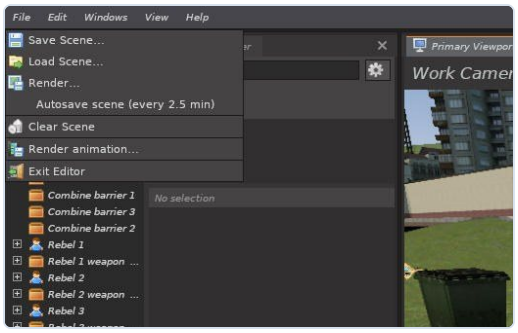
The editor remembers the layout you had when you closed it and restores it next time.

> NEW On screens narrower than about 1430 px (1366×768, 1280×720, 1024×768) the default layout used to break: the left panel covered the viewport and windows became floating. Since a0.4 the layout fits to any screen, and a saved layout that does not fit the current screen is replaced by the default one.

Menus

A reference of every item in the menu bar.

File



Save Scene...

Save the scene under a name.

Load Scene...

Opens the Scene tab with the list of scenes.

Render...

Opens the Render window.

Autosave scene

A backup every 2.5 minutes; several older copies are kept.

Clear Scene

Deletes every object (asks first).

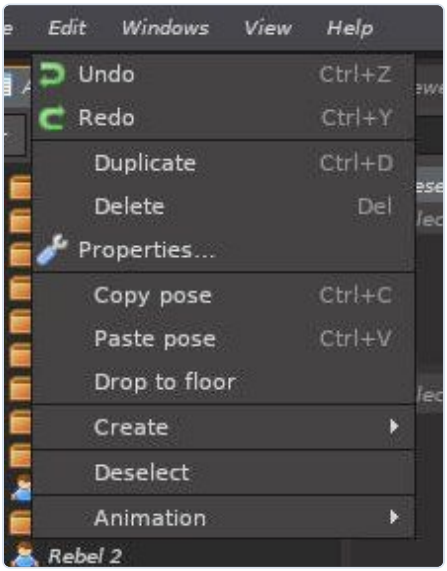
Render animation...

Added by SAeditor: the animation render.

Exit Editor

Closes the editor.

Edit



Undo / Redo

`Ctrl` + `Z` / `Ctrl` + `Y`. Every edit can be undone, including deleting objects.

Duplicate / Delete

`Ctrl` + `D` / `Del` for the selection.

Properties...

The Object Properties window.

Copy / Paste pose

`Ctrl` + `C` / `Ctrl` + `V`, also between different models with the same skeleton.

Drop to floor

Puts the object on the surface below it.

Create

Model, camera, spot or point light, effect, three-point rig.

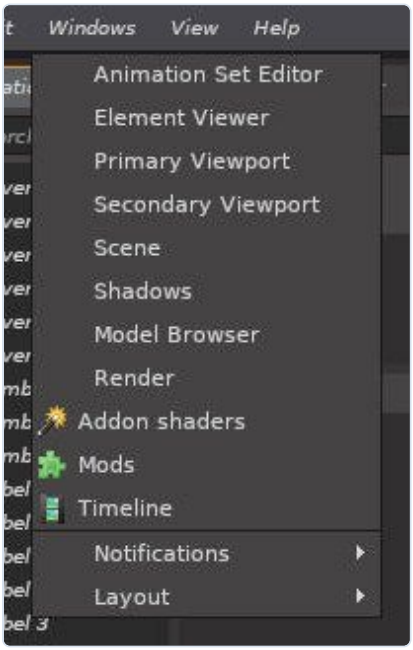
Deselect

Clears the selection.

Animation

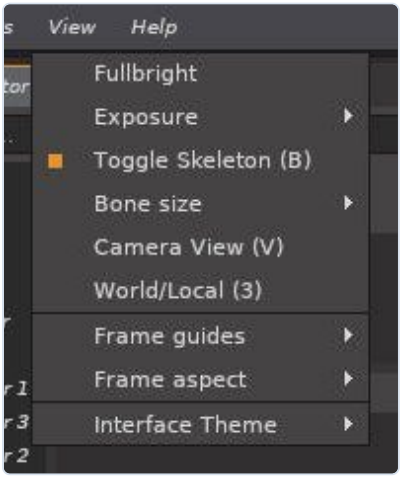
SAeditor commands (keys, import, sounds).

Windows



Shows any window or tab: Animation Set Editor, Element Viewer, both viewports, Scene, Shadows, Model Browser, Render, Addon shaders, Mods, Timeline. **Notifications** sets up the pop-up messages; **Layout** is described on the workspace page.

View



Fullbright

Shows the scene without lighting — handy when building in the dark.

Exposure

Fixed brightness of the viewports or *Auto (engine)*.

Toggle Skeleton (B)

Bone markers on characters.

Bone size

Tiny, Small, Medium, Large markers.

Camera View (V)

Look through the selected scene camera.

World/Local (3)

Gizmo axes in world or object space.

Frame guides

Rule of thirds, center cross, safe areas, golden ratio.

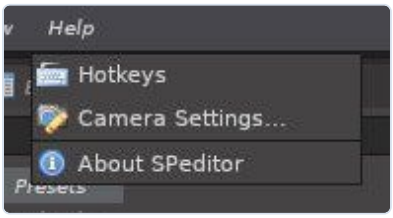
Frame aspect

The shape of the picture (Free, 2:3 poster, A-series and others).

Interface Theme

See Themes.

Help



Hotkeys lists every key (also those of mods), **Camera Settings...** sets the work camera speed and FOV, **About SPeditor** shows the version, the news and the credits.

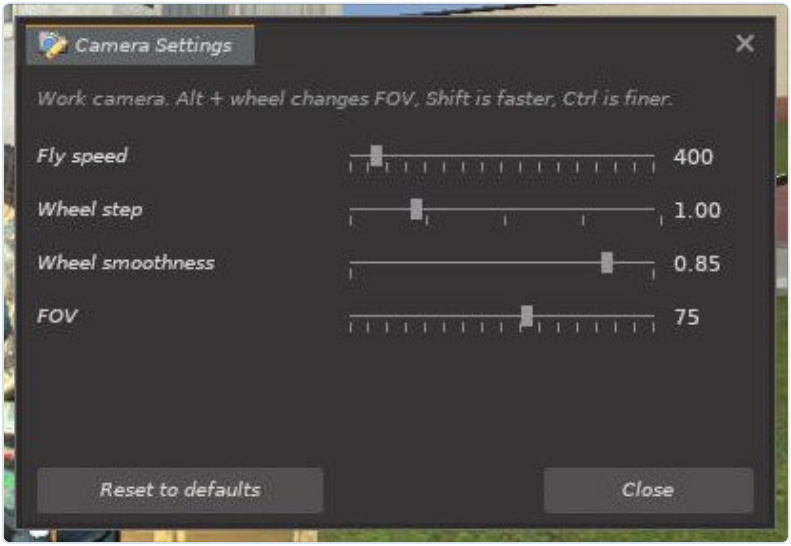
> NEW In a0.3 the keys in the Edit menu were drawn with a square box instead of a gap ("Undo□Ctrl+Z"). Since a0.4 they are aligned on the right, as in the screenshot above.

Viewports and the work camera

You look at the scene through the **work camera** — a free camera that is not part of the scene and never appears in renders.

Moving around

INPUT	ACTION
`RMB` + `W` `A` `S` `D`	Fly. `Q` / `E` go down / up, `Shift` is faster, `Ctrl` slower. Moving the mouse with the right button held turns the camera.
`Alt` + `LMB`	Orbit around the selection.
Wheel	Move forward / back (dolly).
`Alt` + Wheel	Change the field of view (zoom lens).
`F`	Focus on the selected object or bone.
`V`	Look through the selected scene camera; `V` again goes back.



▸ **Help → Camera Settings...** sets the **fly speed**, the **wheel step** and its **smoothness**, and the **FOV** of the work camera. **Reset to defaults** brings back 400 / 1.00 / 0.85 / 75°.

The viewport header and toolbar

The header shows the camera on the left (*Work Camera* or the scene camera's name), the scene name, the frame aspect and size in the middle, and the field of view on the right. The dark bars around the picture are the **frame gate**: only what is inside them gets into the render.

The toolbar under the picture holds the tool buttons (move, rotate, scale, world/local axes, IK), view toggles (frame guides, skeleton, fullbright), the camera list and the gear menu.



The **gear menu** of a viewport: Focus selected, Camera Settings, Render, Exposure, Frame aspect, Hide viewport. The camera list chooses what the viewport shows; in the Secondary Viewport **Auto** shows the selected scene camera, and the work view while you look through a camera in the Primary one.



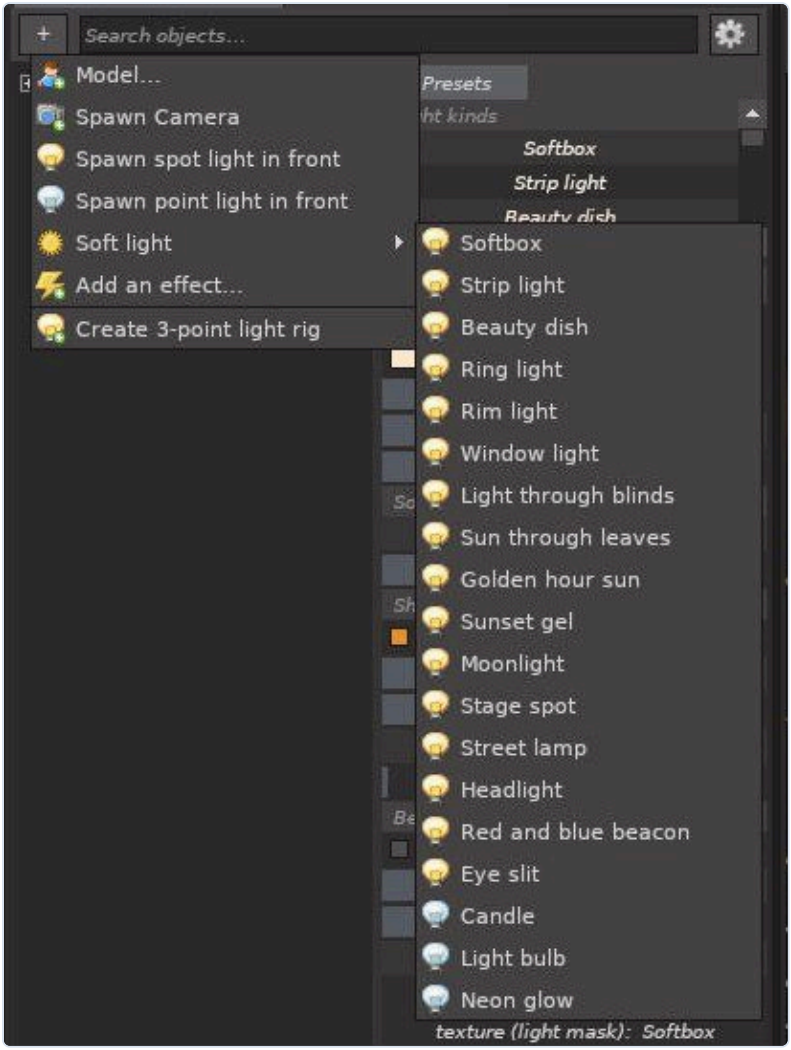
Frame guides help with composition: rule of thirds, a center cross, safe areas and the golden ratio. They are drawn only in the viewport, never in the render.

> ✓ The Primary and Secondary viewports pause while the Render window is open, to give the preview all the speed. The FPS counter reads about 20 when the game window is not focused — that is the engine saving power, not SPeditor.

Objects

A scene is made of four kinds of objects: **models** (props and characters), **lights**, **cameras** and **effects**. All of them are listed in the Animation Set Editor.

Adding objects: the + menu



Model...

Opens the Model Browser.

Spawn Camera

A scene camera exactly where the work camera looks.

Spawn spot / point light in front

A plain light in front of the camera.

Soft light

One of 19 ready-made light kinds (see Lights).

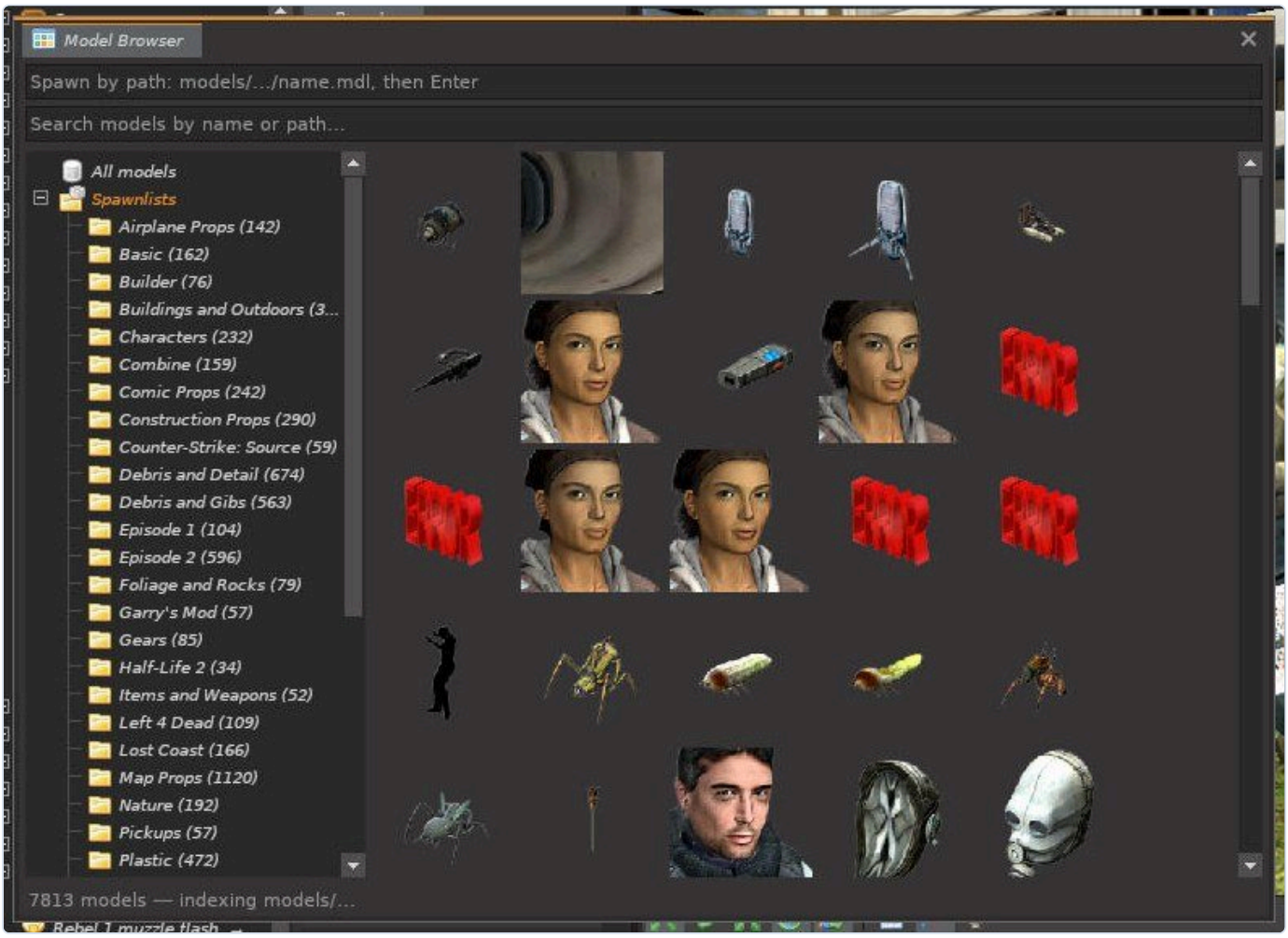
Add an effect...

Opens the Effect Browser (see Effects).

Create 3-point light rig

Key, fill and rim lights around the selected model.

The Model Browser



Model Browser. Folders on the left: *All models* and the game's spawnlists; icons on the right. Click an icon to add the model to the scene.

- **Search** by part of the name or the path, e.g. `alyx`, `combine`, `props_c17`.
- **Spawn by path**: type a full path such as `models/player/group01/male_07.mdl` and press `Enter` — handy for models from addons that are not in any spawnlist.
- Any model works: props, ragdolls, player models and NPC models from the game and from the Workshop. Models with a physics skeleton become posable **ragdolls**; the others are **props** that can still have bones.
- The status line shows how many models are indexed; the list of *All models* grows while indexing runs in the background.

The object tree



The left column of the Animation Set Editor lists every object of the scene. Icons tell the kind: a person for characters, a box for props, a bulb for lights, a camera, a lightning bolt for effects. An attached object shows its parent after an arrow: *Rebel 1 weapon → Rebel 1*.

Expand a character to see its **control groups**: Face, Eyes, Body, Head, RightArm, LeftArm, RightLeg, LeftLeg, RightFingers, LeftFingers. Clicking a group shows only its controls in the right column.

The right column holds the controls of the selection: face presets and flex sliders, eye target, bone rows (*Bone - pos*, *- rot*, *- scale*), or the settings of a light or camera.

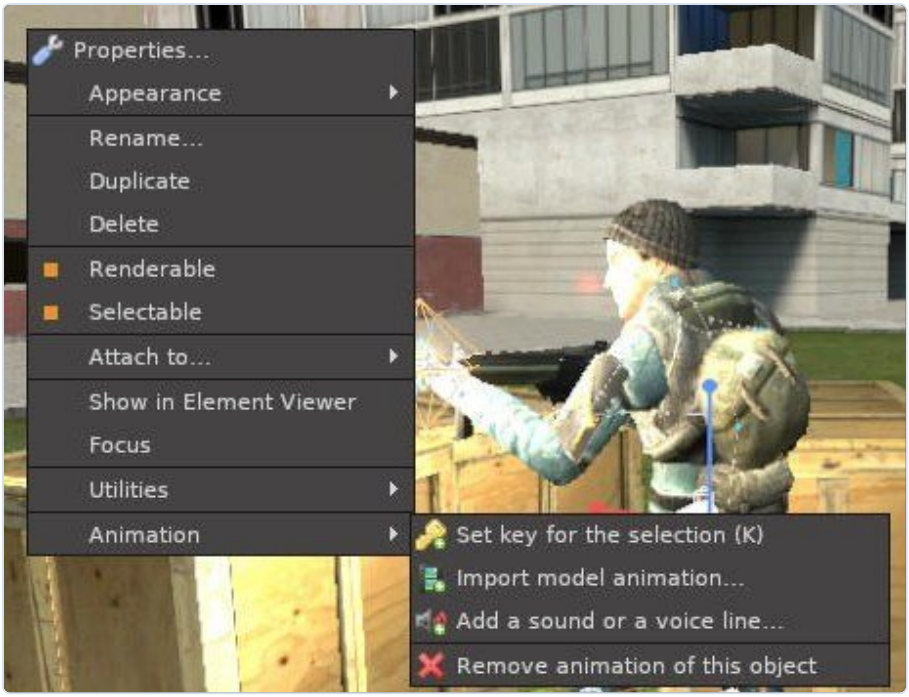
The search field filters the tree. The gear button holds tree options (expand / collapse all).

Selecting

- `LMB` on an object in a viewport or in the tree selects it. `LMB` on a bone marker selects that bone.
- `Ctrl` + `LMB` adds an object to the selection or removes it. Several selected objects are moved, rotated, scaled, duplicated and deleted together.
- Click on empty space or use ▸ **Edit → Deselect** to clear the selection.

The object menu

A short `RMB` click on an object (without moving the mouse) in the viewport or in the tree opens its menu.



Properties...

Colour, textures, size, shadow (below).

Appearance

Quick colour, remove / restore textures, cast shadow, reset.

Rename... / Duplicate / Delete

As named.

Renderable

Off: the object is hidden in viewports and renders.

Selectable

Off: the object is locked and cannot be clicked by accident.

Attach to...

See Attaching and wearing.

Show in Element Viewer / Focus

As named.

Utilities

Drop to floor, 3-point rig, reset pose, copy / paste pose.

Set Skin / Set Body

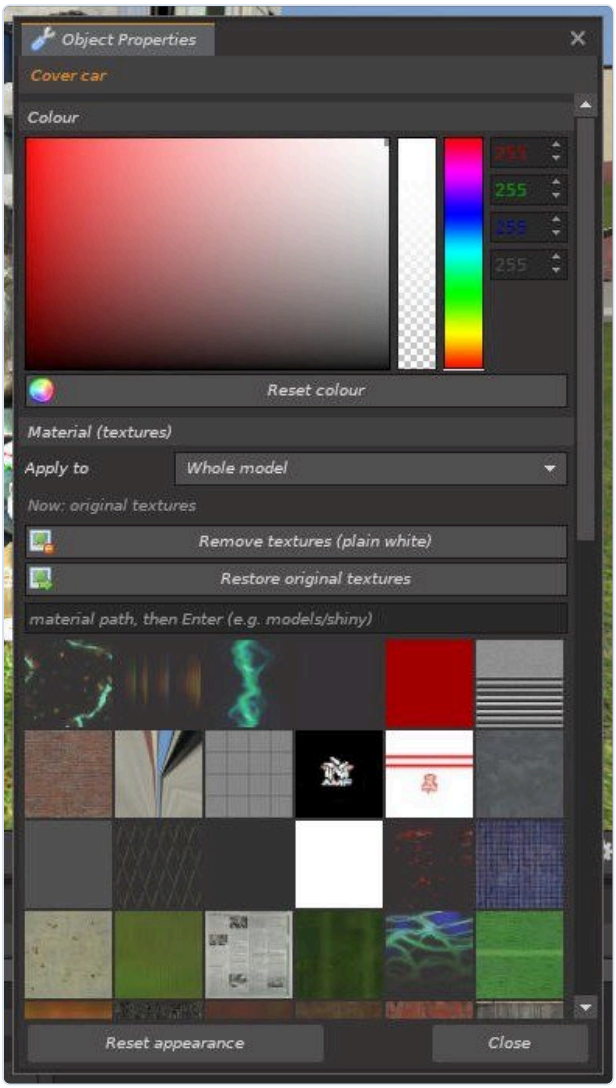
Skins and bodygroups of the model.

Animation

SAeditor: set a key, import an animation, add a sound.

Cameras and lights also have **Aim camera at object...**; cameras have **Look through this camera** and **Match camera to work view**; effects have **Play from the start** and **Choose another effect...**.

Object Properties



Colour

Tint and opacity (the fourth field is alpha: below 255 the object becomes see-through). *Reset colour* returns white.

Material

Replace textures for the *Whole model* or one material slot. Pick a material from the grid or type a path such as `models/shiny`. *Remove textures* makes the model plain white so it takes the object colour; *Restore original textures* undoes it.

Size

Props only: overall scale and stretching in length (X), width (Y) and height (Z). The Scale tool `4` does the same with the mouse.

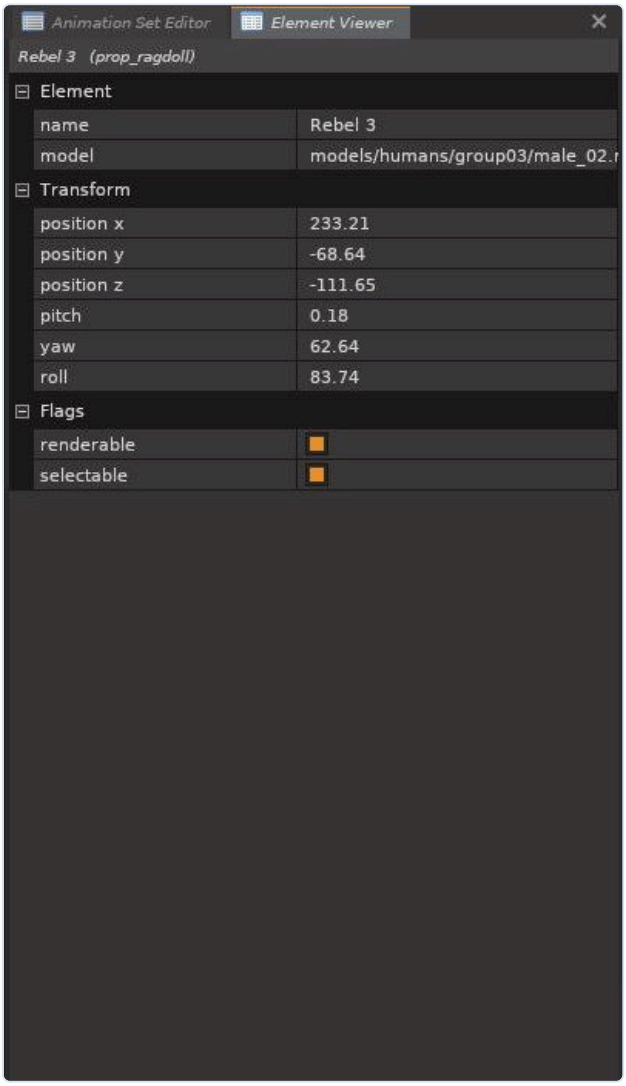
Shadow

Cast shadow off removes the object from shadow maps — useful for an object in front of a lamp.

Reset appearance

Everything back to the original.

Element Viewer



The Element Viewer tab (next to the Animation Set Editor) shows every property of the selection as numbers: name, model, position x/y/z, pitch/yaw/roll, scale, skin, renderable, selectable, and for lights and cameras all their settings with SFM names (*intensity, fieldOfView, castsShadows, shadowFilterSize…*). Type a value for exact placement.

> ✓ Any slider in SPeditor: `Shift` + drag is fine, `Ctrl` + drag is coarse; right-click a slider for *Reset to default* and *Enter value…*.

Posing

Everything is posed with one gizmo: whole objects and single bones, including fingers.

The gizmo



Move 1



Rotate 2

KEY	TOOL
1	**Move.** Drag an arrow to move along an axis, the centre to move in the screen plane.
2	**Rotate.** Drag a ring to turn around its axis.
4	**Scale.** The centre scales evenly, an axis stretches along it. Works on props and on bones of any model.
3	**World / local axes.** Local axes follow the object's own rotation.
I	**Inverse kinematics.** Move a hand or a foot and the whole arm or leg follows.

Bones



The left hand selected with IK on. The small white cubes are bone markers.

Click a bone marker to select the bone, or click a row *Bone - pos / - rot / - scale* in the Animation Set Editor: it selects the bone and the matching tool at once.

Bones with a physics body (pelvis, spine, head, upper arms, forearms, hands, thighs, calves, feet) move the ragdoll itself. Bones without one (clavicles, fingers, toes, face bones) are moved with bone offsets. Both are saved in poses and animated by SAeditor.

With ▸ **View → Toggle Skeleton (B)** the markers are hidden; ▸ **View → Bone size** makes them larger or smaller.

Scaling bones NEW

Select any bone and press `4` (or click its - *scale* row): a big head, long arms, a giant hand. In a0.3 the Scale tool worked only on bones without physics (fingers) and silently acted as Move on a hand or the head; a0.4 scales every bone.

Face and eyes

- **Face presets** at the top: *Default, Zero, Half, One* set all flexes at once.
- **Flex sliders** are grouped as Brows, Eyes, Nose, Mouth, Other (e.g. *smile, jaw_drop, blink*). Models without face flexes do not show the section.
- **Eyes:** tick *viewTarget* to aim the eyes; *eyes_rightleft*, *eyes_updown* and *eyes_distance* set where they look.

- **Skins and bodygroups**: object menu → ▸ **Set Skin** / ▸ **Set Body**.

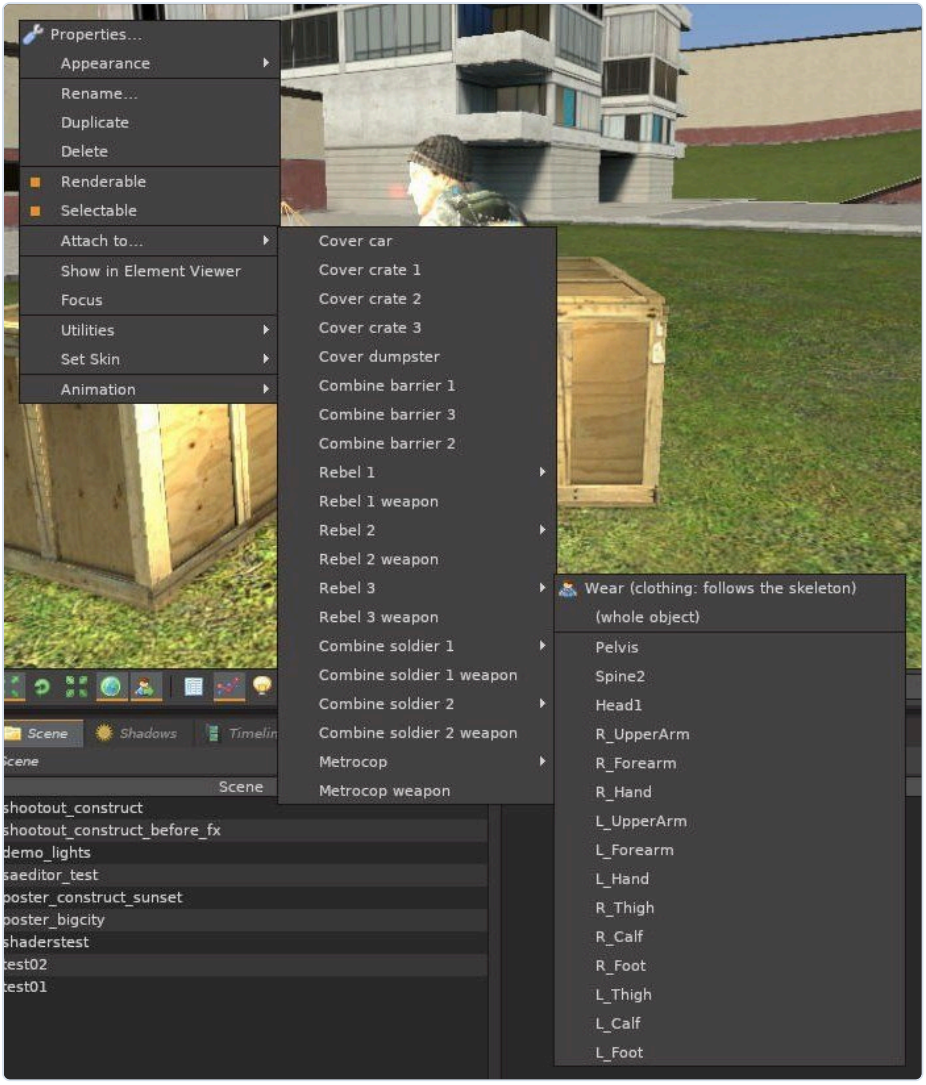
Poses: copy, paste, library

- `Ctrl`+`C` / `Ctrl`+`V` copy the pose of the selected character and paste it onto another. Between different models with the same skeleton the pose is carried over by bone names, keeping each model's proportions.
- The **Pose library** in the Scene tab: *Save pose*, *Load pose*, *Delete*. The list shows the pose name and the model it was made on.
- ▸ **Utilities** → **Reset pose of selected** returns bones, face and eyes to the original.

> Addons that change ragdolls (active ragdoll systems, Fedhoria-like mods) can fight SPeditor and make limbs impossible to pose. If limbs jump back or fall, disable such addons while you work in the editor.

Attaching and wearing

Objects can follow other objects: a weapon in a hand, a hat on a head, a muzzle flash on a gun, or clothing that bends with the body.



Object menu → Attach to... → a character. At the top *Wear*, then *(whole object)*, then every physics bone.

Three ways to attach

CHOICE	RESULT	USE FOR
R_Hand, **Head1**...	Follows that bone rigidly, keeping its offset.	Weapons, props in hands, hats, helmets as separate props.
(whole object)	Follows the object's root (pelvis of a character).	A rider on a vehicle, a light on a car.

Wear NEW	Every bone of the worn model follows the bone with the same name on the wearer (bone merge).	Clothing, armour, hair, accessories made for the same skeleton (e.g. ValveBiped).

Putting clothes on a character

1. Add the clothing model (from a clothing addon) and the character.
2. Right-click the clothing → ▸ **Attach to...** → the character → ▸ **Wear (clothing: follows the skeleton)**.
3. Pose the character: the clothing bends with every limb. Skin, bodygroups and colour of the clothing still work.
4. To take it off: object menu → ▸ **Detach**.

> **i** Before a0.4 clothing could only be attached rigidly to one bone, so it did not follow the arms and legs. *Wear* arrives in a0.4; it is saved with the scene, survives duplicating and can be undone.

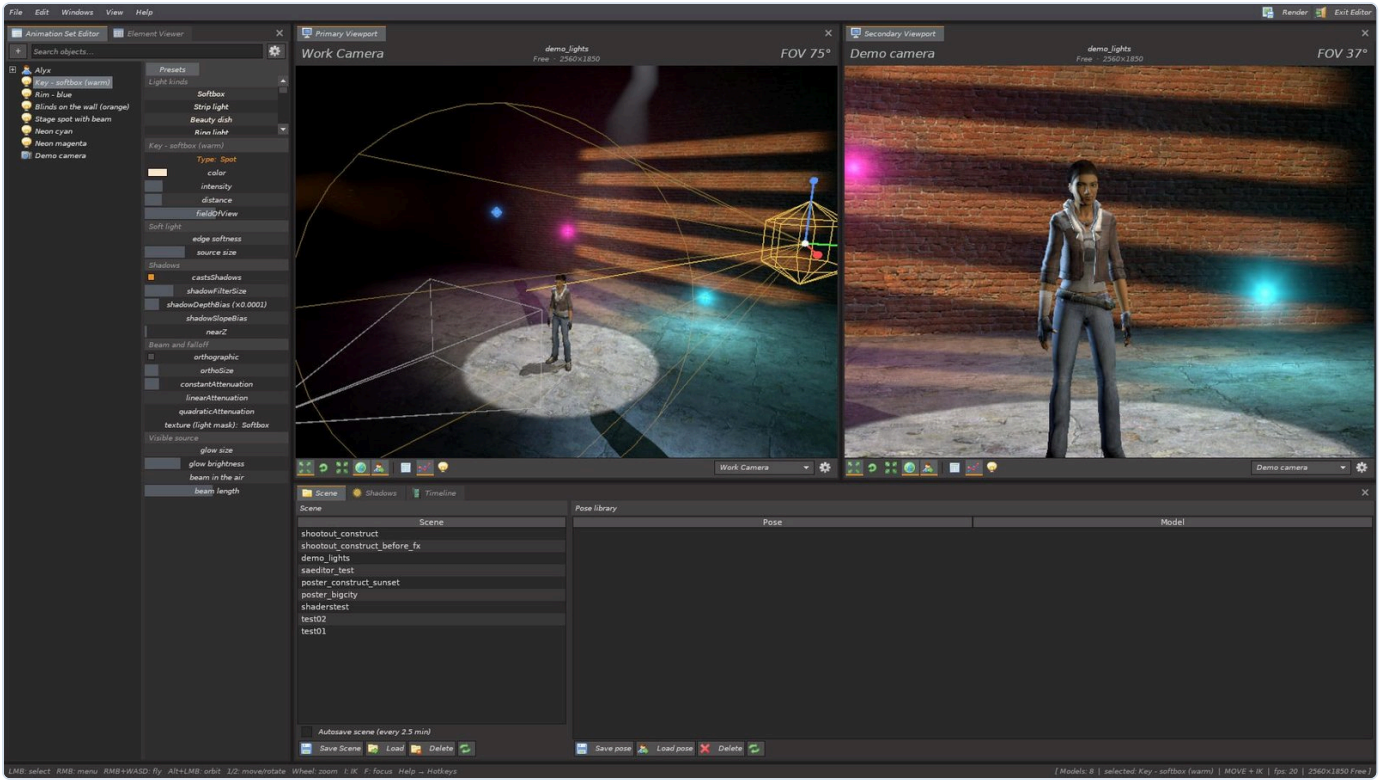
> **!** Wearing works only when the bone names match. A model made for another skeleton (TF2, a custom rig) stays at the wearer's root. You can still click and select the worn object: its invisible copy stays where it was attached.

Rules

- Move an attached object by hand and the new offset is kept.
- Chains are allowed (a scope on a gun in a hand); an object cannot be attached to its own child.
- When the parent is deleted, the object stays where it was.
- Attachments are saved in scenes and followed in SAeditor animations.

Lights

Light makes a poster. SPeditor has point lights and spot lights; spot lights cast shadows and have a soft edge, penumbra, a glow and a visible beam.



Six lights: a warm softbox as the key light, a blue rim light, orange “light through blinds” on the wall, a stage spot with a beam, two neon glows. The selected softbox shows its cone and its controls in the Animation Set Editor.

Point and spot

	POINT LIGHT	SPOT LIGHT
Shines	in all directions	in a cone or through a mask
Shadows	no (an engine limit)	yes, soft, adjustable
Good for	fill light, neon, fire, bulbs, muzzle flashes	key and rim light, sun, windows, lamps

Switch a light between the two kinds with its *Type* control in the Animation Set Editor.

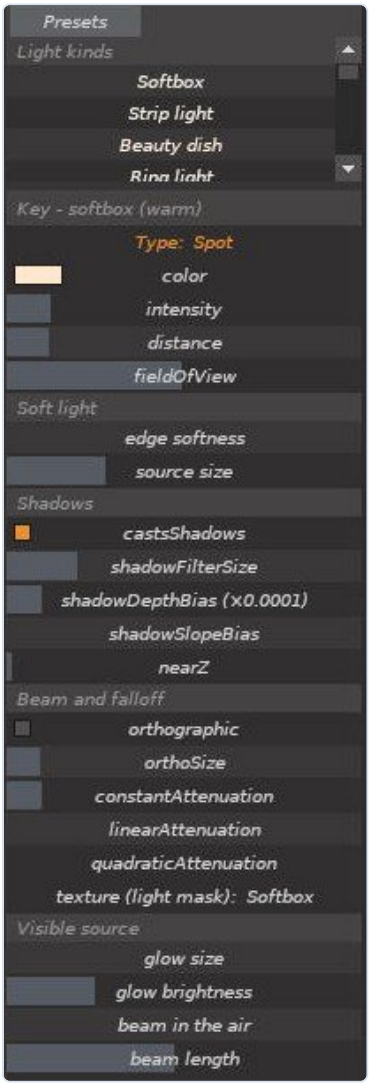
Light kinds

+ → ▸ **Soft light** creates a light already set up for a purpose. With a light selected, the *Light kinds* list at the top of the right column turns that light into another kind.

KIND	WHAT IT IS
Softbox	Large soft key light, the studio standard.
Strip light	A narrow tall softbox for edges and reflections.
Beauty dish	Round, contrasty portrait light.
Ring light	Ring around the lens: flat face light, round catchlights.
Rim light	Back light that outlines the silhouette.
Window light	Daylight through a window frame.
Light through blinds	Stripes of light, the noir look.
Sun through leaves	Dappled sunlight.
Golden hour sun	Low warm sun with long shadows.
Sunset gel	A coloured gradient of sunset light.
Moonlight	Cold dim parallel light.
Stage spot	Hard round spot with a visible beam.
Street lamp	Warm cone from above.
Headlight	Car headlight with a glow.
Red and blue beacon	Police lights.
Eye slit	A narrow strip of light across the eyes.
Candle · Light bulb · Neon glow	Small point sources with a visible glow.

Presets (colour only): Soft white, Warm lamp, Cool moonlight, Neon cyan, Neon magenta, Fire.
Create 3-point light rig places key, fill and rim lights around the selected model.

Light settings



color · intensity

Colour (click the swatch for a palette and a mixer) and brightness.

distance · fieldOfView

How far the light reaches; the cone angle of a spot.

edge softness

0 keeps the mask as it is, 1 fades the light from the centre to the edge.

source size

Size of the lamp: a larger source gives softer shadows that blur with distance (penumbra). Calculated in the render and its preview.

castsShadows · shadowFilterSize

Shadows on/off and the blur of their edge.

shadowDepthBias · shadowSlopeBias

Raise if lit surfaces get stripes; lower if the shadow detaches from the object.

nearZ

The light starts at this distance from the source.

orthographic · orthoSize

A parallel square beam like sunlight: shadows keep their size at any distance.

constant / linear / quadratic attenuation

Falloff with distance. Quadratic is like a real lamp; constant does not fade at all (good for far subjects).

texture (light mask)

The shape of the beam: soft masks of SPeditor (softbox, strip, round, ring, window, blinds, bars, slit, leaves, sunset...), Lamp tool textures, other game textures or any texture by path.

glow size · glow brightness

A visible glow around the source, like a headlight or a beacon.

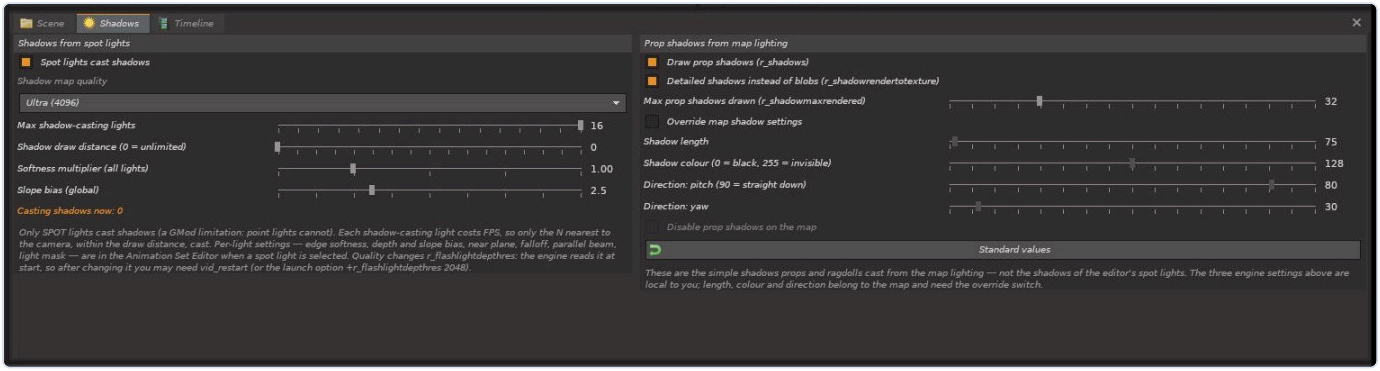
beam in the air · beam length

A visible cone of light, as in dust or fog.

Every shadow-casting spot light costs FPS. The Shadows tab limits how many of them cast shadows at once. Penumbra (*source size*) is only calculated in the render; the viewports show a sharp shadow.

Shadows

The Shadows tab of the bottom window controls two different things: shadows of the editor's spot lights and the simple prop shadows the map's own lighting casts.



Shadows from spot lights

Spot lights cast shadows

The master switch.

Shadow map quality

Low, Medium, High, Ultra. The engine reads it at start: after a change you may need `vid_restart` or the launch option `+r_flashlightdepthres 2048`.

Max shadow-casting lights

Only the N lights nearest to the camera cast shadows. The line below shows how many cast now.

Shadow draw distance

A light whose lit area is farther than this casts no shadow (0 = unlimited). Saves FPS in large scenes.

Softness multiplier

Multiplies the edge blur of every light.

Slope bias (global)

Removes stripes on surfaces lit at a shallow angle; each light can override it.

Prop shadows from map lighting

These are the soft blobs or silhouettes that props and ragdolls drop from the map's sun. *Draw prop shadows*, *Detailed shadows instead of blobs* and *Max prop shadows drawn* are local engine settings. Length, colour and direction belong to the map: tick *Override map shadow settings* to

change them (this affects every player on a server; untick to restore). *Disable prop shadows on the map* removes them entirely.

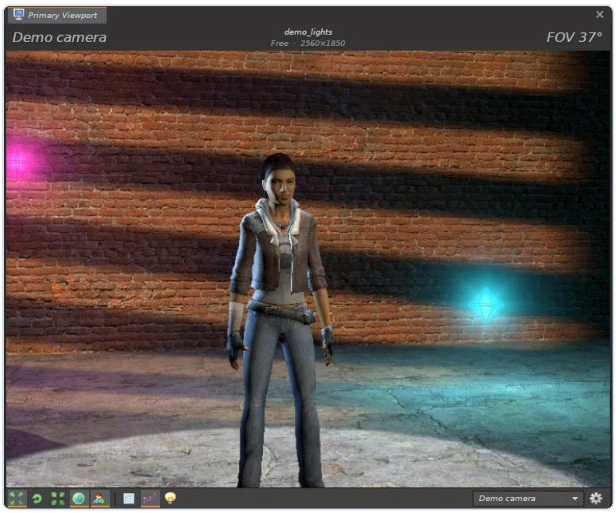
Cameras

Scene cameras are objects of the scene. A render is always taken through a scene camera (or through the Primary Viewport), and camera effects belong to scene cameras.

Creating and aiming

- **+** → ▸ **Spawn Camera** creates a camera exactly at the work camera's position and angle.
- **Match camera to work view** (object menu) moves an existing camera to where you look now.
- **Aim camera at object...** turns it towards any object.
- Move and rotate it with the gizmo like any object.

Looking through a camera



Select a camera and press `V`, choose it in the viewport's camera list, or tick *Look through this camera*. The header then shows the camera name. Flying with `RMB` + `WASD` now moves the scene camera itself — the fastest way to frame a shot. `V` goes back to the work camera.

Camera settings



Lenses

Ready focal lengths: 18 mm (90°), 24 mm (74°), 35 mm (54°), 50 mm (40°) and longer.

fieldOfView

Any angle.

Shake, units / degrees

How far the camera is thrown about and turned. A small value gives a handheld camera.

Shake speed

Shakes per second: 1–2 handheld, 8+ an explosion or an engine.

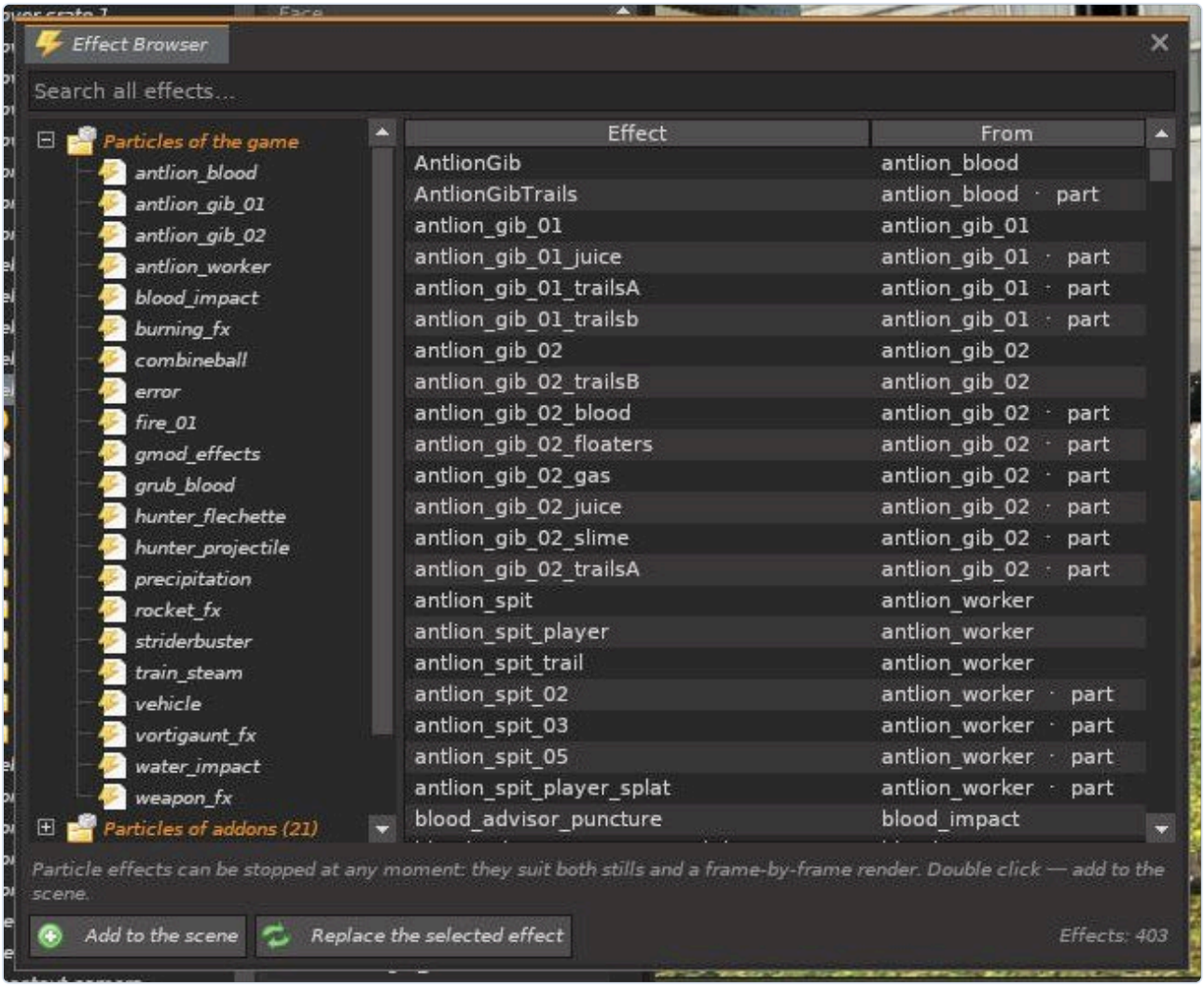
Smoothness, s

In an animation the camera floats through its keys instead of stopping at each one.

A camera's own **depth of field** (focus distance, aperture, quality) and **exposure**, as in SFM, are set in the Render window under *Camera effects → Camera (as in SFM)*. See Rendering.

Effects

Explosions, fire, smoke, blood, sparks and muzzle flashes are objects too. A particle effect can be stopped at any moment and stays frozen in the viewport and in the render.



Effect Browser (**+** → ▸ **Add an effect...**). Groups on the left, effects and the file they come from on the right. Double-click adds the effect to the scene.

Kinds of effects

GROUP	CAN BE FROZEN	NOTES
Particles of the game / of addons	✓	The best choice for stills and frame-by-frame video. Includes every particle file of installed Workshop addons.

Emitter tool effects	—	Play by themselves; a still render starts them and waits for the chosen moment.
Engine effects · Lua effects	—	Some need a weapon or a character and show nothing alone.

Effect settings

Moment, s

How long after its start the effect is shown. 0 means it has not started yet. After a move or a change of the moment the effect is replayed up to that moment, which takes as long as the moment itself.

Size

Scale of the effect (not every effect uses it).

Length

Distance to the second point: where a tracer or a beam goes. 0 means a one-point effect.

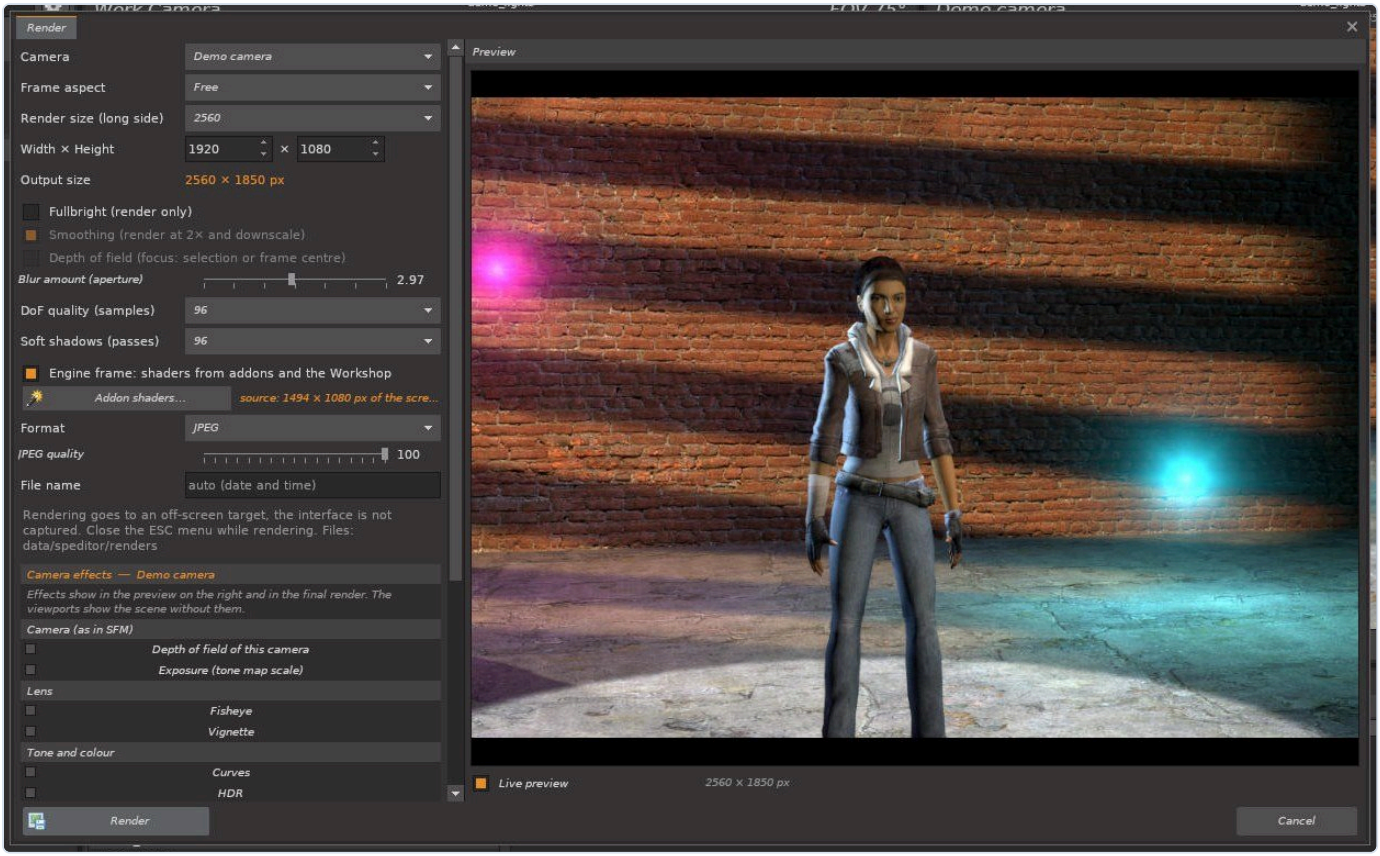
Effects can be attached to an object or a bone (a flash on a muzzle), duplicated, hidden, aimed and saved with the scene. In SAeditor, **Play the effect from this frame** keys the moment for you.

> An effect advances one game frame at a time. Render animations with the game window focused: an unfocused game runs at about 20 FPS, and effects shorter than a frame (a muzzle flash lasts ~0.03 s) may not show.

Rendering

The Render window turns the scene into a picture. It renders off-screen, so the interface never gets into the file, and shows a live preview of exactly what you will get.

Open it with the ▸ **Render** button at the top right, ▸ **File → Render...** or the viewport gear menu.



The Render window for a still scene. Settings on the left, the preview on the right. The preview grows with the window.

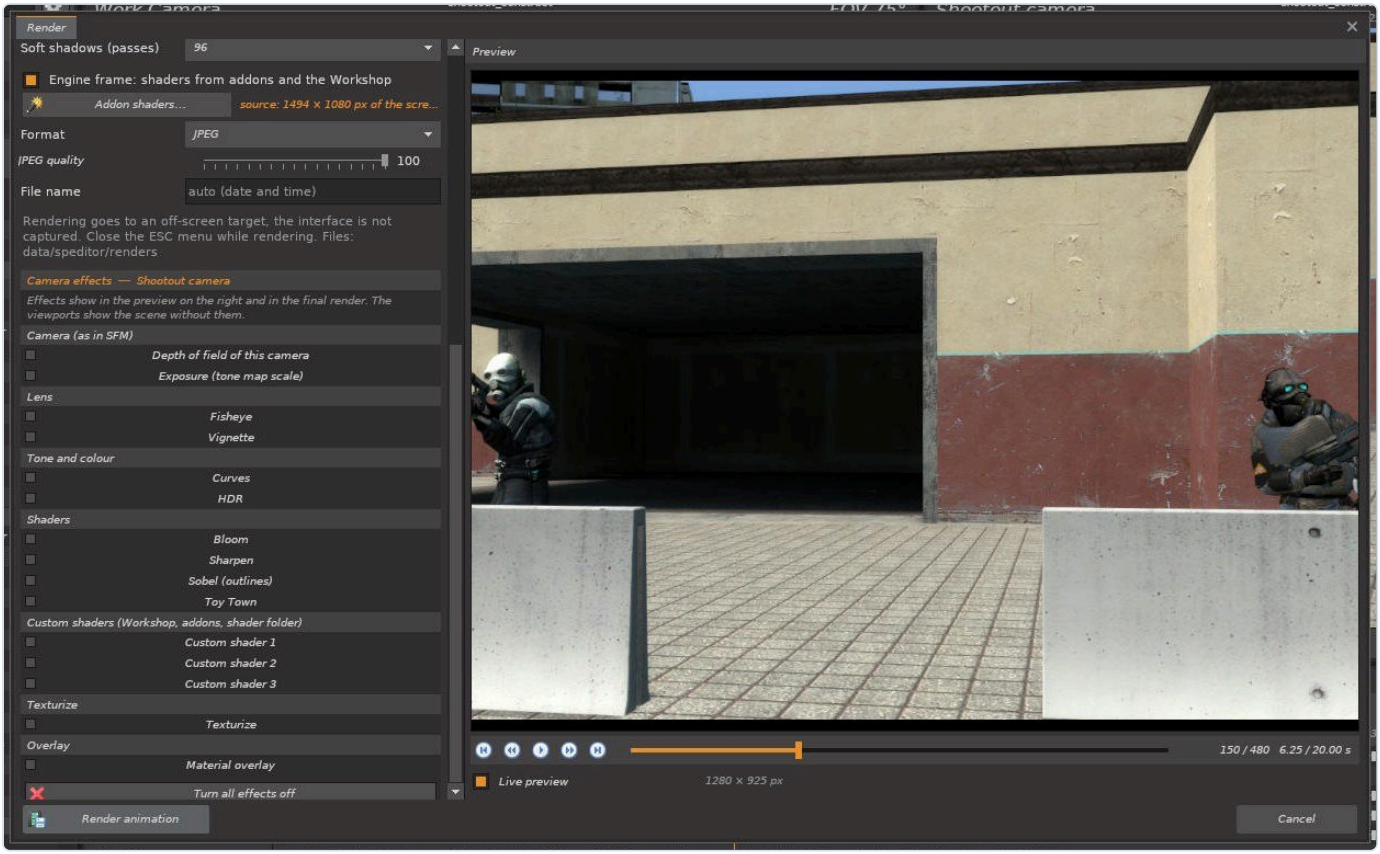
Settings

SETTING	MEANING
Camera	A scene camera, or *Primary Viewport source* (whatever the Primary Viewport shows).
Frame aspect	Free (the viewport's shape), 2:3 poster, A-series paper and other ratios.
Render size (long side)	The long side of the picture, up to 3840 px. *Custom size* sets width × height exactly. *Output size* shows the result.
Fullbright (render only)	Render without lighting.

Smoothing	Renders at 2× and scales down: clean edges.
Depth of field	Focus on the selection or the frame centre; *Blur amount* (aperture) and *DoF quality* (passes).
Soft shadows (passes)	How many times the scene is rendered for lights with a *source size*. More passes, smoother penumbra, longer render.
Engine frame	Lets the game's own renderer draw the picture so that post-processing addons and Workshop shaders get in. See Shaders.
Format · JPEG quality	PNG (lossless) or JPEG.
File name	Empty means date and time. Files go to `data/speditor/renders/`.

> **!** Close the ESC menu while rendering. A large render with depth of field and soft shadows can take a while — the preview shows the progress (“DoF preview: 12 of 96 samples”).

Camera effects

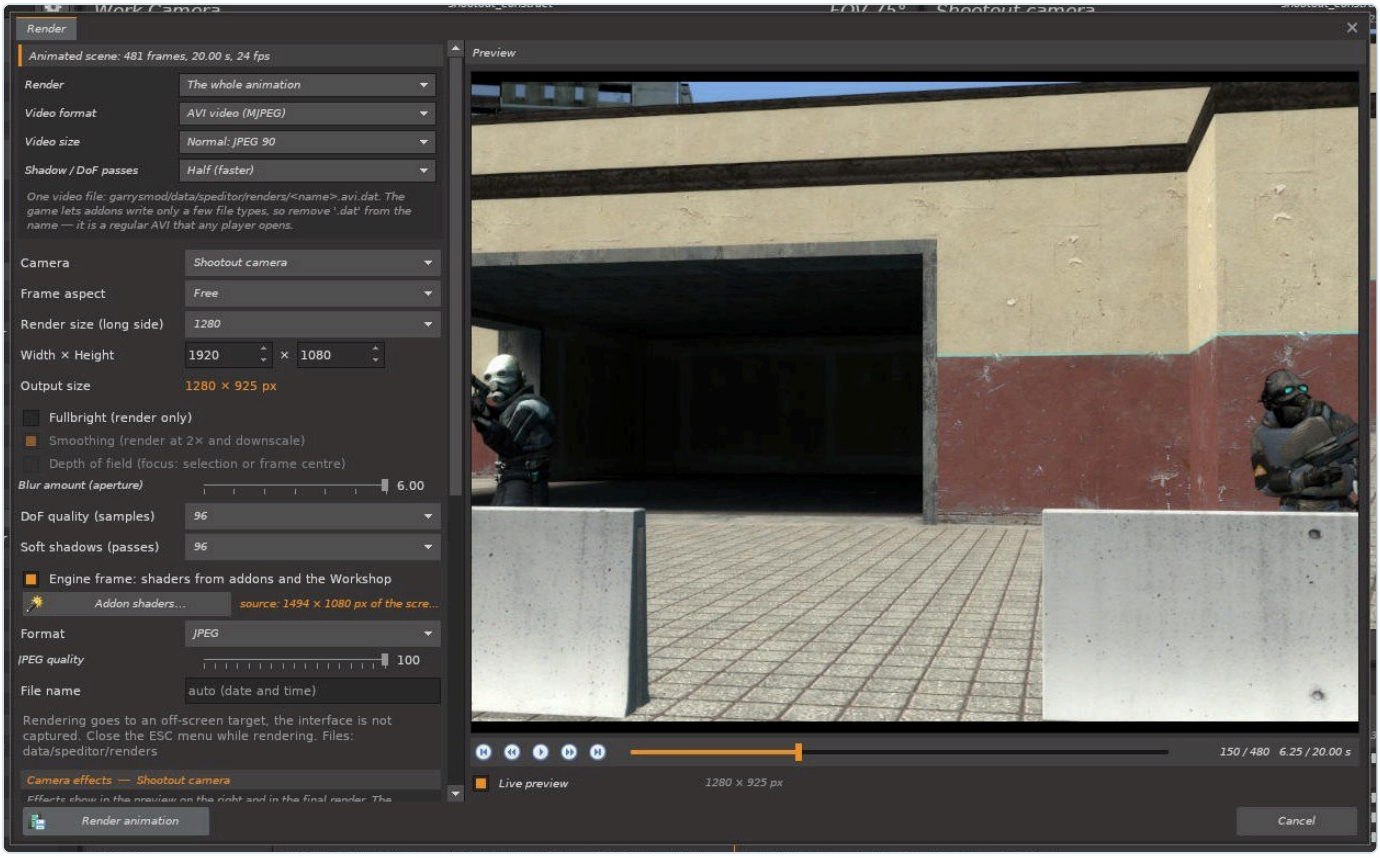


Camera effects at the bottom of the settings. They belong to the chosen scene camera and are saved with it.

SECTION	EFFECTS
Camera (as in SFM)	Depth of field of this camera (focus distance, aperture, quality) · Exposure (tone map scale)
Lens	Fisheye (strength) · Vignette (strength, clear radius)
Tone and colour	Curves for RGB and each channel (click to add a point, drag to move, right-click to remove) · HDR
Shaders	Bloom · Sharpen · Sobel (outlines) · Toy Town
Custom shaders	Up to three shader materials from the Workshop, addons or the shader folder
Texturize · Overlay	A texture pattern over the picture · any material over the picture

The effects show in the preview and in the final render; the viewports stay clean. The work camera has no effects — choose a scene camera. *Turn all effects off* resets them.

Animated scenes



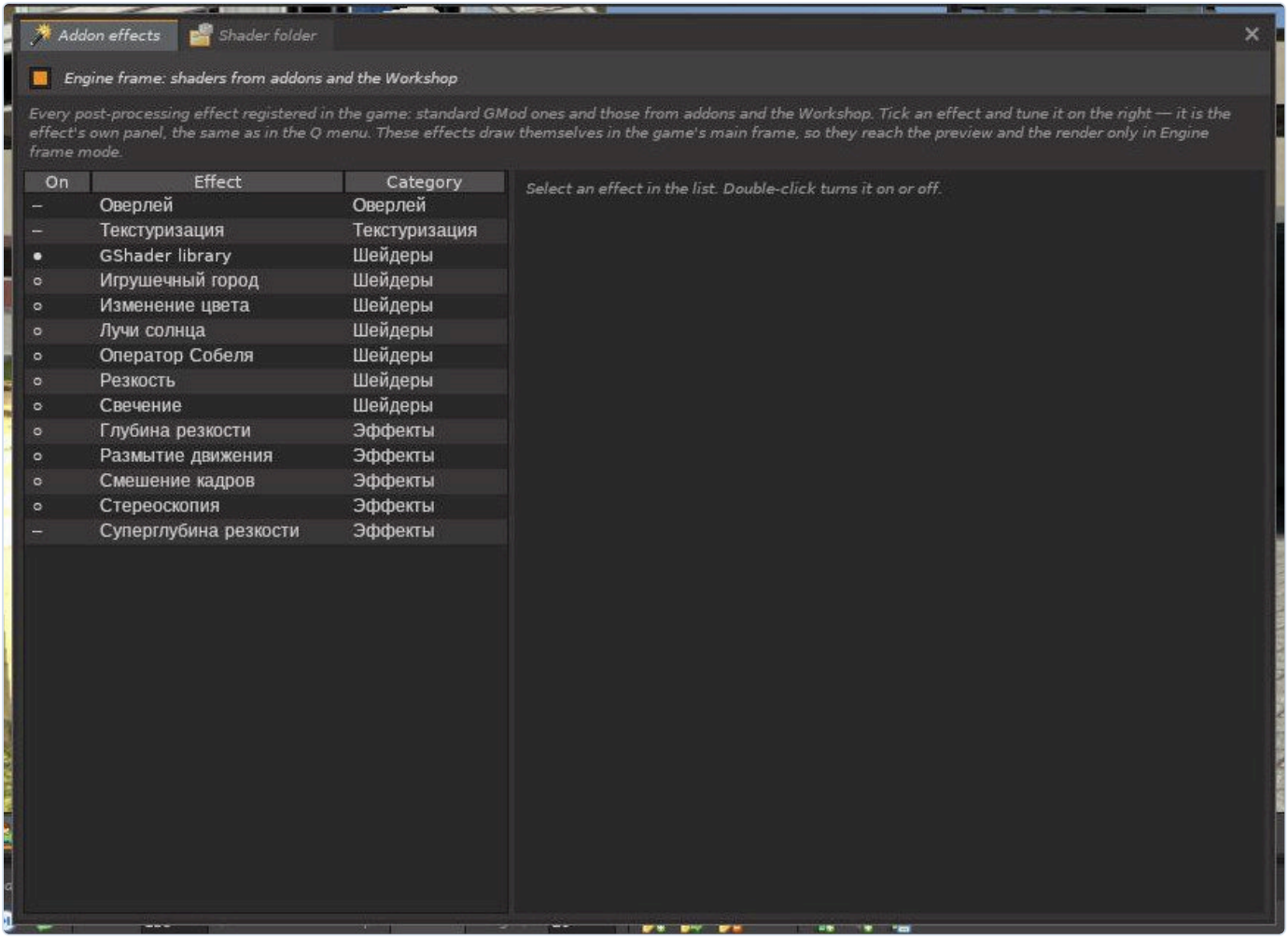
When the scene has an animation the window starts with SAeditor settings: whole animation or the current frame, video format, video size and passes, and a player under the preview.

See Animation → Video for the formats.

✓ `speditor_render` in the console renders with the current settings — bind it to a key for quick test shots.

Workshop shaders

Post-processing addons and shader packs from the Workshop can get into your renders in two ways: through **Engine frame** or as **custom shaders**.



Windows → Addon shaders. Every post-processing effect registered in the game, with its category. Double-click turns an effect on or off; its own settings panel appears on the right.

Engine frame

Tick *Engine frame* in the Render window. The picture is then drawn by the game's renderer through the chosen camera with every enabled effect — standard GMod effects, Workshop shaders and shaders built on GShader Library — and taken from the screen.

> **!** Engine frame limits: the picture is no larger than the screen, and 2× smoothing, penumbra and the editor's depth of field are not applied. The Render window shows the source size, e.g. “source: 1494 × 1080 px of the screen”.

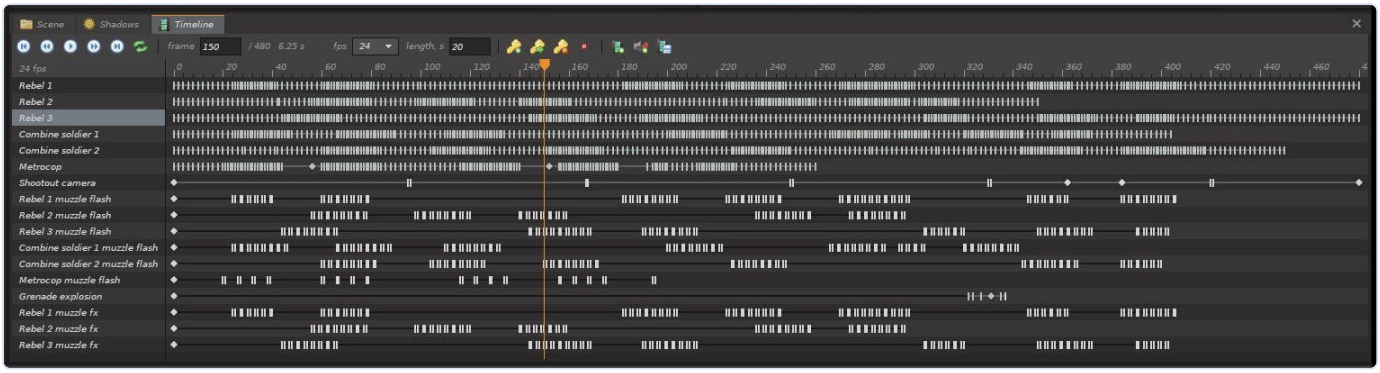
The shader folder

The *Shader folder* tab of the same window: put a downloaded shader pack (with `materials/` and `shaders/fxc/`, or loose `.vmt` and `.vcs` files) into `garrysmod/addons/speditor/shaders/` (create the folder if it is missing) and press *Rescan*. The pack is mounted without restarting the game, and its shaders appear in *Camera effects* → *Custom shader*. Lua from a pack is not run: addons with Lua are installed the usual way and work through Engine frame.

> On Linux the folder name and the files inside must be lowercase, otherwise the game cannot read them.

Animation: keys and the timeline

SAeditor is the animation mod that comes with SPeditor. It adds the **Timeline** tab to the bottom window. You set keys — full snapshots of an object at a frame — and SAeditor calculates the motion between them.



The timeline of the example scene. One row per animated object. Diamonds are keys; dense marks are baked animations (a key every frame or two). The orange line is the playhead.

Your first animation

1. Select an object, leave the playhead on frame 0 and press `K`. A key appears in the object's row.
2. Move the playhead (click the ruler or press `.` several times), change the object — move it, change the pose, the light colour, the camera angle — and press `K` again.
3. Press `Space` to play. Between the keys everything is interpolated.

A key remembers everything about the object: position, physics pose, bone offsets (scale too), face flexes, colour and opacity, light settings, camera FOV and shake, the moment of an effect, and NEW skin and bodygroups (they switch at the key instead of blending).

The toolbar

CONTROL	ACTION
Transport buttons	First frame (`Home`), previous key, play / pause (`Space`), next key, last frame (`End`), loop playback.
frame	Current frame; type a number to jump. Next to it: total frames and time.
fps · length, s	Frame rate (12, 15, 24, 30, 60) and length in seconds (up to 3600 frames).

Key buttons	Set key for the selection (`K`), set key for every object, delete key, **Auto key** (the red dot: every edit sets a key at the current frame; the status bar shows AUTO KEY).
Right buttons	Import model animation, add a sound, render animation.

Working with keys

- `LMB` on a key selects it and moves the playhead there; drag to move it; `Ctrl` + drag copies it.
- `RMB` on a key: *Motion to the next key* — **Smooth** (ease in and out), **Linear** or **Hold** (jump: the value holds until the next key, used for cuts between shots); *Copy key*; *Delete key*.
- `RMB` on an empty place of a row: *Set key here*, *Paste key here*, *Remove animation of this object*.
- `LMB` on the ruler or empty space moves the playhead; drag to scrub. The wheel scrolls the list of objects.

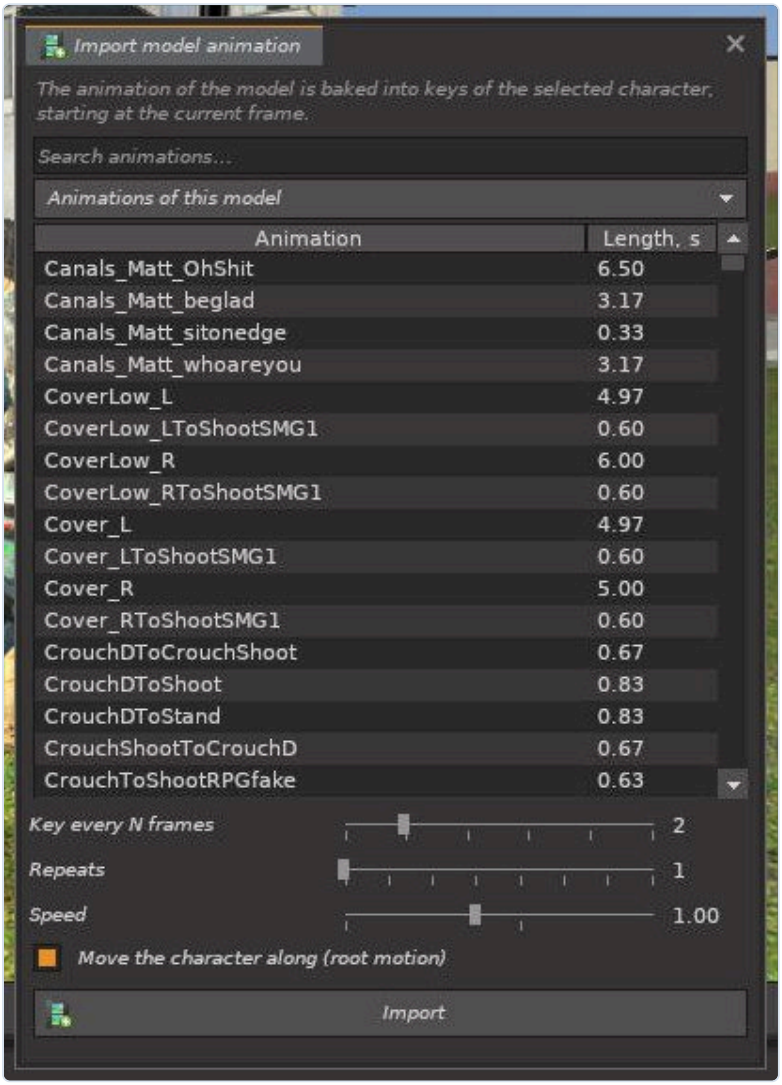
KEY	ACTION
`Space`	Play / pause
`K`	Set a key for the selection
`,` / `.`	Previous / next frame
`Home` / `End`	First / last frame

> The animation is stored with the scene: ▸ **File → Save Scene...** saves it too. Physics poses are set on the server, so playback is smooth in singleplayer and on a listen server; on a busy dedicated server it depends on the network.

Importing model animations

A character can get a ready-made animation — walking, running, gestures, dances — baked into keys. Right-click a character →

▸ **Animation → Import model animation...**



Source list

Animations of this model, *Player animations (male / female / zombie)* or *Animations of another model...* (type a path). Any model with the same skeleton works: a citizen can dance like a player model or use an animation pack made for players.

Search

Filter by name; the list shows each animation's length.

Key every N frames

1 is the most exact; 2–3 is lighter for long animations.

Repeats

Loop a walk cycle several times.

Speed

Play faster or slower.

Move the character along

Root motion: a walk moves the character forward instead of walking in place.

The animation starts at the current frame. If it is longer than the timeline, the timeline is lengthened. Select several characters first to give all of them the same animation — they move in step.

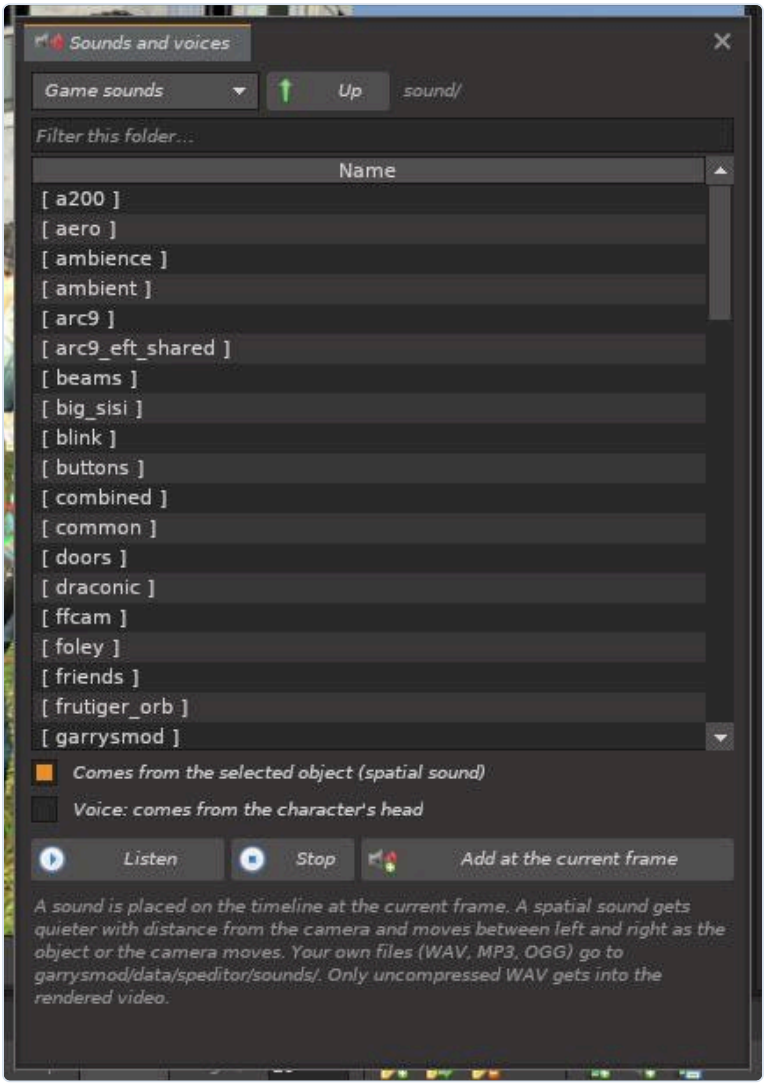
Fingers NEW

In a0.3 only bones with a physics body were imported, so fists, pointing and gripping hands stayed open. Since a0.4 the import also brings fingers, toes and clavicles as bone offsets: a fist animation clenches the fist.

> “This model has another skeleton” means the source model's bones do not match the character. Choose a source made for the same skeleton (most humans in GMod use ValveBiped).

Sounds and voices

Sounds are placed on the timeline and played with the animation. A sound can come from an object (spatial), from a character's head (voice) or be heard everywhere (music, a narrator).



Open it with
▸ **Animation → Add a sound or a voice line...**
or the speaker button of the timeline.

- **Game sounds** browses the `sound/` folders of the game and addons; **My files** shows `garrysmod/data/speditor/sounds/` (WAV, MP3, OGG).
- **Listen** / **Stop** preview a file.
- **Comes from the selected object**: quieter with distance from the camera, moves between left and right.
- **Voice**: comes from the character's head (ticked by itself for files in `sound/vo/`).
- **Add at the current frame** or double-click a file.

Sounds on the timeline

Every sound is a blue bar under the objects; the same file from the same source shares one row (a burst of shots is one row). Drag a bar to move it in time. Right-click it for:

Listen · Comes from · Voice

Preview; change the source object or make it heard everywhere; toggle voice.

Volume · Speed and pitch · Distance of full volume

10–200 %, 50–200 %, 50–2400 units.

Start at the current frame · Duplicate · Remove the sound

As named.

Cutting and trimming NEW

- **Drag the left or right edge** of a bar (the cursor becomes ↔) to cut off the start or the end of the sound. A trimmed side is marked with a light stripe.
- Menu: *Cut in two at the playhead*, *Trim the start to the playhead*, *Trim the end to the playhead*, *Restore the whole sound*.
- Trimming works in playback and in the video's sound track; a few milliseconds of fade at each cut prevent clicks.

> Only uncompressed WAV gets into the rendered video. MP3 and OGG play in the editor but are left out of the video (the render tells you which ones). Convert them to WAV first.

Rendering an animation to video

When the scene has an animation, the Render window offers *The whole animation* or *Only the current frame*, and a video format.

FORMAT	RESULT
AVI video (MJPEG)	One file `renders/<name>.avi.dat` with sound. Remove `.dat` from the name — it is a regular AVI any player opens. A video over 1.9 GB continues in a second file.
JPEG frames	Numbered frames in `renders/<name>/` plus a WAV sound track — for making MP4.
PNG frames	Lossless frames for a video editor.

Video size

Five JPEG levels: Largest (100), Normal (90), smaller by 40 % (70), by 65 % (50), by 80 % (30). Frames that did not change are not written twice.

Shadow / DoF passes

All (as in a still), Half or Quarter — fewer passes render several times faster.

Size, smoothing, depth of field and camera effects come from the same Render window. A progress line appears on the timeline; *Stop the animation render* cancels it.

Making an MP4

Garry's Mod cannot write MP4 itself. After the render the console prints a ready ffmpeg command. For an AVI:

```
ffmpeg -i myclip.avi -c:v libx264 -crf 18 -pix_fmt yuv420p -c:a aac myclip.mp4
```

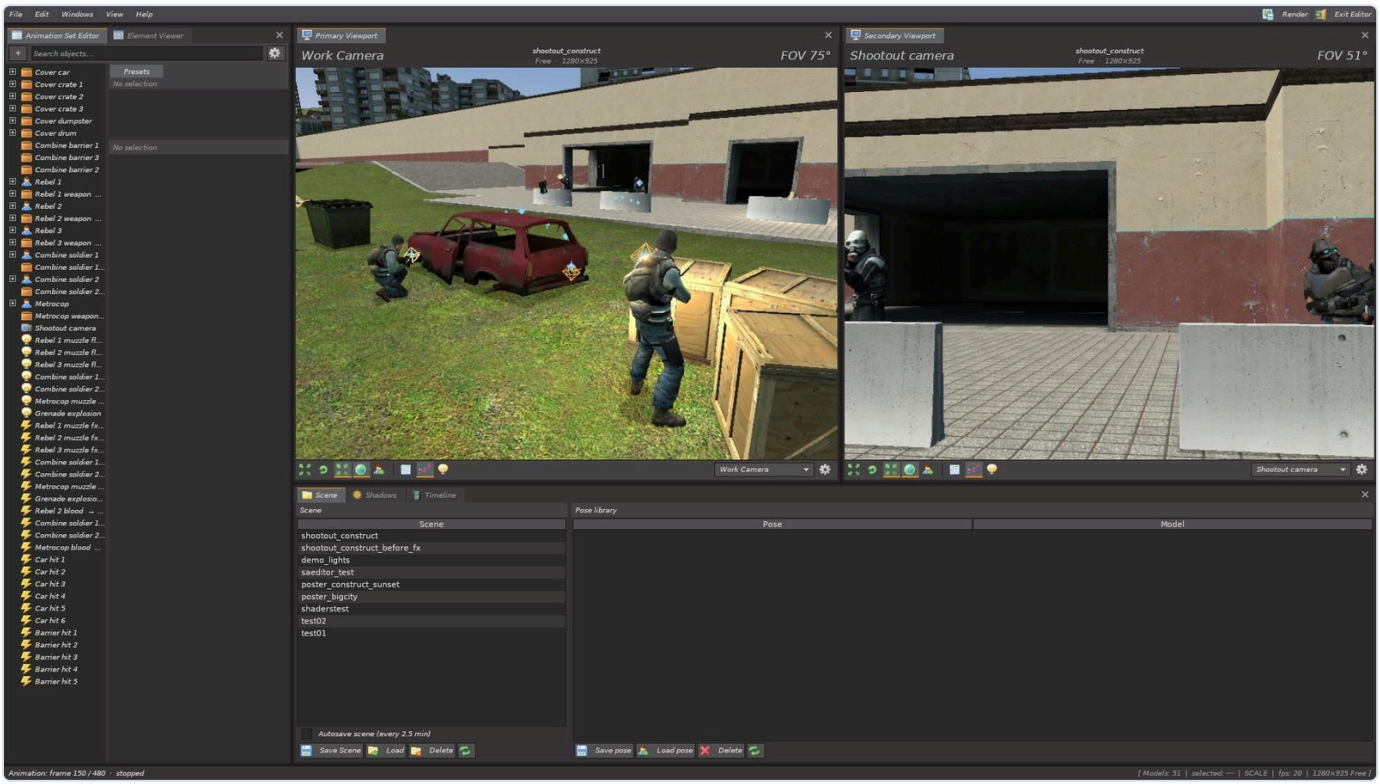
For JPEG frames (24 fps):

```
ffmpeg -framerate 24 -i myclip_%04d.jpg -i myclip.wav -pix_fmt yuv420p myclip.mp4
```

> **!** Keep the game window focused while an animation renders: unfocused, the game runs at about 20 FPS, which slows the render and can skip very short effects.

Scenes, poses and undo

The Scene tab of the bottom window keeps your work between sessions.



Bottom window, Scene tab: the scene list on the left, the pose library on the right.

Scenes

- **Save Scene** asks for a name. **Load** replaces the current scene (asks first). **Delete** removes the file. The refresh button rereads the list.
- A scene keeps: every object with its pose, face, appearance and attachments; light settings; camera settings and camera effects; the work camera; the frame settings (aspect, size, exposure, depth of field); the SAeditor animation and sounds; data of other mods.
- The map is recorded too. Loading on another map warns you: coordinates would put objects in random places.
- **Autosave scene (every 2.5 min)**: `autosave` is the newest copy, `autosave_2` ... are older ones.
- `shootout_construct` is an example that comes with the addon; it cannot be deleted.

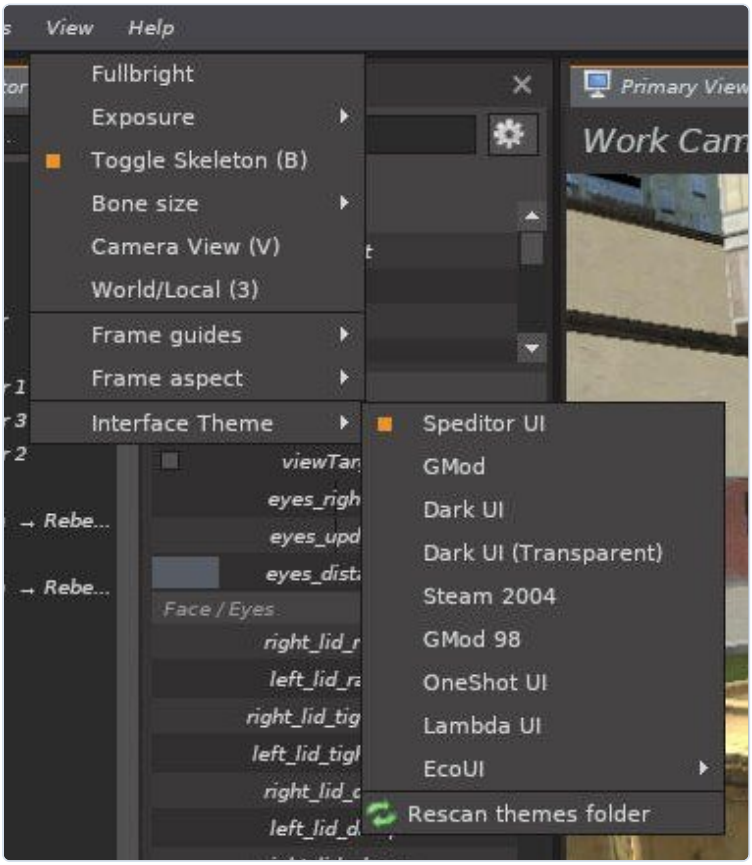
Undo and redo

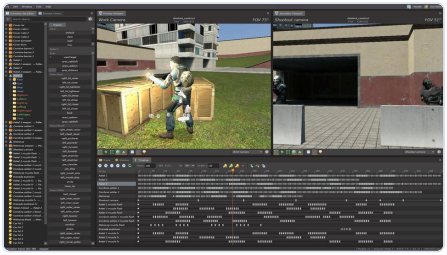
`Ctrl` + `Z` / `Ctrl` + `Y` work for every edit: moving, posing, face, lights, cameras, appearance, attaching, keys and sounds of SAeditor, even deleting objects (a deleted object waits in a trash until the history forgets it).

Themes and notifications

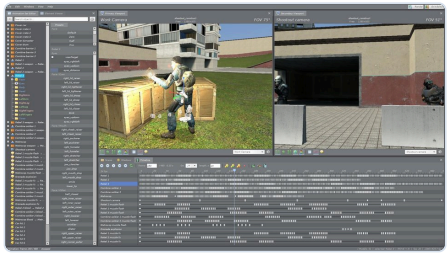
Interface themes

▸ **View → Interface Theme** or Q menu → SPeditor → **Theme**. Changing the theme keeps open windows, the selected camera and fullbright. The editor uses only its own copies of the skins, so interface addons from the Workshop cannot break it.

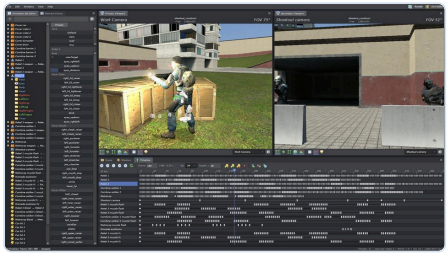




Speditor UI



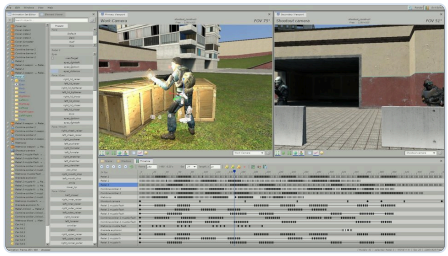
GMod



Dark UI



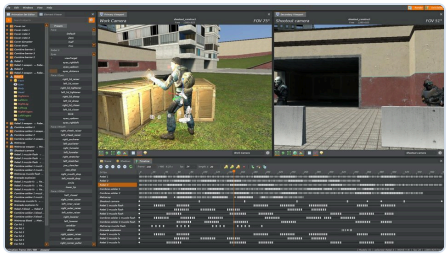
Steam 2004



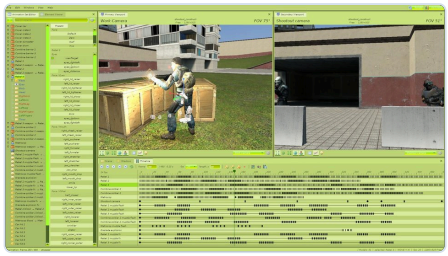
GMod 98



OneShot UI



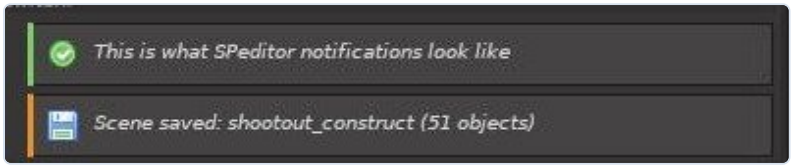
Lambda UI



EcoUI (16 colour variants)

There is also *Dark UI (Transparent)*. Your own Derma skin: put it as a folder into `garrysmod/addons/speditor/themes/` (create the folder if it is missing; a Lua file with `derma.DefineSkin`) and use *Rescan themes folder*.

Notifications



Saves, autosaves, renders, errors and hints appear as cards in the bottom right corner. ▸ **Windows → Notifications**: show pop-ups, sound, display time, the *Message log* with every message of the session, and a test message.

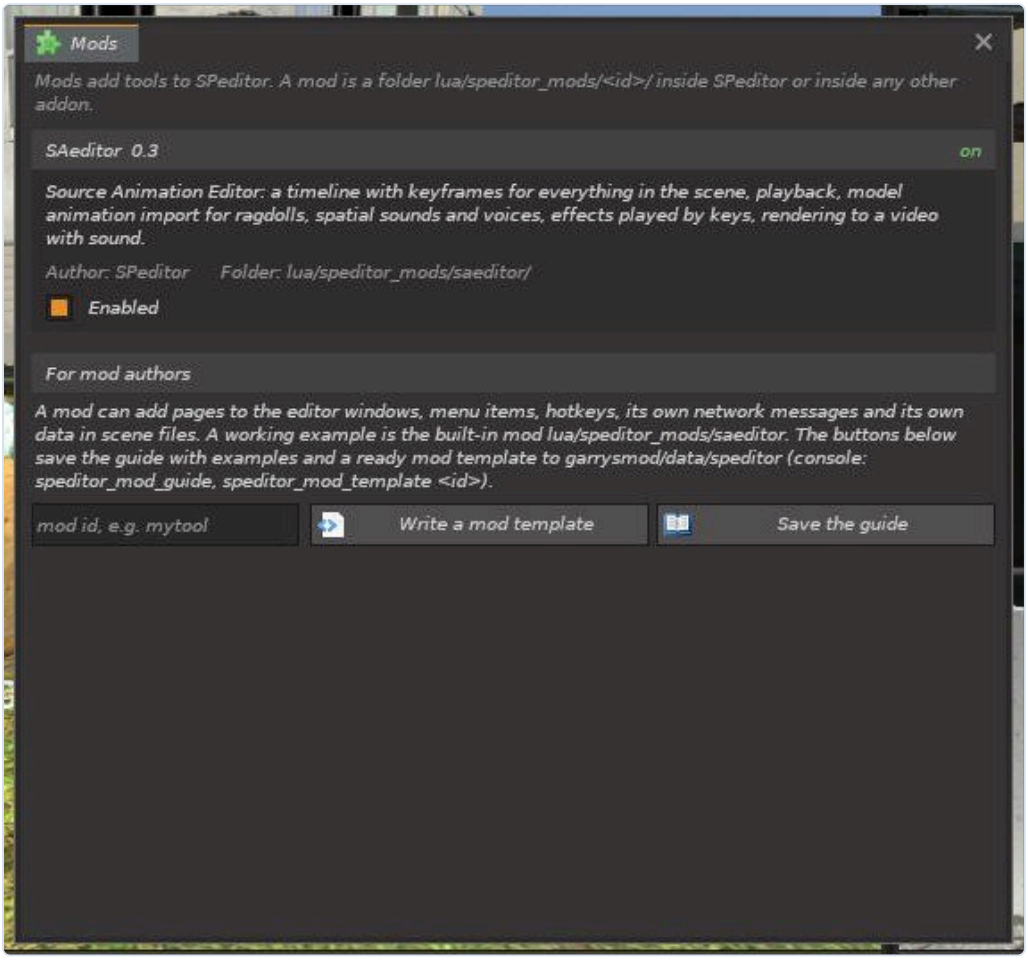
About



▸ **Help → About SPeditor**: the version, what is new in it, the list of features and the authors of the bundled skins (click a name to open its Workshop page, right-click to copy the link).

Mods

A mod adds tools to SPeditor: pages in the editor windows, menu items, hotkeys, its own network messages and its own data in scene files. SAeditor itself is a mod and the working example.



▸ **Windows → Mods** lists installed mods with their state (on, off, error, needs a newer SPeditor), author and folder. Untick *Enabled* to switch a mod off (applied after the map restarts).

For mod authors

1. A mod is a folder `lua/speditor_mods/<id>/` with a `mod.lua`, inside SPeditor or inside any other addon.
2. **Save the guide** writes the full guide with examples to `garrysmod/data/speditor/` (console: `speditor_mod_guide`).
3. Type an id (Latin letters, digits and `_`, not starting with a digit) and press **Write a mod template** (console: `speditor_mod_template <id>`) for a ready skeleton to start from.

Hotkeys

The same list is in the editor: ▸ **Help** → **Hotkeys**. Mods add their keys to it.

KEY	ACTION
SELECTION AND CAMERA	
`LMB`	Select object / bone
`Ctrl` + `LMB`	Add to / remove from selection
`RMB` (short click)	Object menu
`RMB` + `W` `A` `S` `D` / `Q` `E`	Fly the work camera (Q down, E up, Shift faster, Ctrl slower)
`Alt` + `LMB`	Orbit around the selection
Wheel	Move the camera in / out
`Alt` + Wheel	Field of view (zoom lens)
`F`	Focus on selection
`V`	Look through the selected camera / back
`B`	Show / hide skeleton
TOOLS	
`1` / `2` / `4`	Move / rotate / scale
`3`	World / local axes
`I`	Inverse kinematics
`Shift` + drag	Fine slider drag (Ctrl: coarse)
EDITING	
`Ctrl` + `Z` / `Ctrl` + `Y`	Undo / redo
`Ctrl` + `D`	Duplicate
`Del`	Delete
`Ctrl` + `C` / `Ctrl` + `V`	Copy / paste pose
WINDOWS	
Drag tab bar	Dock: drop on a window edge / tab bar (Ctrl: keep floating)

Double-click tab bar	Maximize / restore window
Right-click tab bar	Window properties
SAEDITOR	
`Space`	Play / pause the animation
`K`	Set a key for the selection
`,` / `.`	Previous / next frame
`Home` / `End`	First / last frame



Help → **Hotkeys** in the editor.

Console commands and settings

Everything here can also be set in the interface; the console is useful for binds and for server owners.

Commands

COMMAND	ACTION
`speditor_toggle`	Open or close the editor
`speditor_render`	Render with the current settings
`speditor_mod_guide`	Save the mod guide to data/speditor
`speditor_mod_template <id>`	Write a mod template

Settings (convars)

VARIABLE	ME
`speditor_access`	Ser `ad` `su` `n` |
`speditor_language`	`en`
`speditor_theme`	Inte the (sp gm dar ste gm one lam eco
`speditor_autosave`	Aut eve min (0/1

`speditor_aspect`	Fra asp
`speditor_render_side` · `_custom` · `_w` · `_h`	Ren the side cus wid hei
`speditor_render_format` · `_quality`	png JPE
`speditor_render_ss` · `_fullbright` · `_preview`	2× sm full ren pre
`speditor_render_engine`	Eng fra
`speditor_dof` · `_aperture` · `_samples`	Dep fiel am pas
`speditor_softlight_samples`	Pas soft sha
`speditor_exposure`	Vie exp = a
`speditor_guides` · `speditor_bone_size`	Fra gui bor ma
`speditor_cam_speed` · `_wheel` · `_wheel_smooth` · `_fov`	Wo can set

`speditor_shadows` · `speditor_shadow_quality` · `_max` · `_dist` · `_softscale` · `_slope`	The Sha tab
`speditor_mapshadow*`	Map sha ove len col pitc off
`speditor_notify` · `_sound` · `_time`	Not
`speditor_mods_off`	Mo swi
`saeditor_render_mode` · `_format` · `_quality` · `_samples` · `_settle`	Ani ren wh fra jpe vid pas lon for sce

Problems and answers

Collected from the questions players ask in the Workshop comments.

The interface is huge / windows overlap on my screen

On screens narrower than ~1430 px the default layout of a0.3 broke. Fixed in a0.4. Until then: ▸ **Windows** → **Layout** → **No bottom panel** or arrange the windows yourself and *Save layout as…*.

Clothes I attach do not move with the model

Use ▸ **Attach to…** → **character** → **Wear** (a0.4). Attaching to a bone is rigid by design: good for weapons, not for clothing. See Attaching and wearing.

How do I resize a bone (a bigger hand or head)?

Select the bone, press `4` and drag the centre (even) or an axis (stretch). In a0.3 this works on fingers and clavicles; a0.4 allows every bone. See Posing.

Can I pose bones without the physgun?

Yes, that is what the editor is for: click a bone marker and use the gizmo, or turn on IK (`I`) and drag a hand or foot.

Imported animation has no finger movement

Fixed in a0.4: fingers are imported as bone offsets. Re-import the animation after updating.

Limbs cannot be moved / snap back

Usually an addon that controls ragdolls (active ragdolls, Fedhoria and similar). Disable it while you use the editor.

Do custom models and Workshop ragdolls work?

Yes, since the first alpha. Find them in the Model Browser or type the path into *Spawn by path*. Models made as SFM “ragdolls” but ported as props can still be posed bone by bone with the gizmo.

Where are my scenes and renders?

In `garrysmod/data/speditor/`. See Where your files are.

Can I save the animation as a video?

Yes: Render window → *The whole animation* → *AVI video*. Remove `.dat` from the file name. For MP4 use the ffmpeg command printed in the console. See Video.

Can I cut a sound?

Since a0.4: drag the edges of a sound bar or use *Cut in two at the playhead*. See Sounds.

Can skins and bodygroups change during an animation?

Since a0.4 they are part of every key and switch exactly at the key.

Does it work with soft lamps?

SPeditor has its own soft lights since a0.2: *source size* gives a real penumbra in the render, plus edge softness and 19 ready light kinds. See Lights.

My shadows have stripes or float away from objects

Stripes: raise *shadowDepthBias* or *shadowSlopeBias* of the light. Floating shadow: lower them. Only spot lights cast shadows.

Workshop shaders do not appear in the render

Tick *Engine frame* in the Render window and turn the effect on in ▸ **Windows → Addon shaders**.

Is it compatible with Stop Motion Helper, Portal Gun, StormFox?

Not tested yet. Please report what you find in the Workshop discussions.

What's new

a0.4 NEW

Made from players' feedback in the Workshop comments.

- **Wear:** clothing, armour and hair made for the same skeleton follow every bone of the character (more).
- **Scale any bone:** hands, head, arms — not only fingers (more).
- **Fingers in imported animations:** fists clench, hands grip (more).
- **Skins and bodygroups in animation keys** (more).
- **Cutting and trimming sounds** on the timeline, in playback and in the video (more).
- Fixed: the layout broke on screens narrower than ~1430 px; a saved layout that does not fit is replaced by the default.
- Fixed: keys in the Edit menu were drawn with a square box.

a0.3

- Effects: particle effects of the game and of Workshop addons, frozen at any moment; Emitter and engine effects.
- Effects in SAeditor: the moment is keyed; “Play the effect from this frame”.
- Sounds and voices on the timeline; video with sound; faster animation render; five video sizes.
- Animations from another model with the same skeleton; several characters at once.
- Camera motion: shake and smoothness. Camera depth of field and exposure as in SFM.
- Example scene `shootout_construct`.

a0.2

- Soft light: source size (penumbra), edge softness, glow, visible beam, 19 light kinds, light masks. The mod system and SAeditor.

Credits

SPeditor is made for Garry's Mod (Facepunch Studios). Source Filmmaker is a product of Valve; SPeditor is not affiliated with Valve. The GMod theme is a copy of the standard Derma skin by Garry Newman. The bundled interface skins belong to their authors:

SKIN	AUTHOR
EcoUI	ELuminumG24
Dark UI Spawnmenu	AquaRex
Steam 2004 Derma Skin	Maks1mDiamond
GMod98 UI Skin	AwesomeGraczGie21
OneShot - UI Skin	Flarky
Lambda UI	□□□□□

Workshop shader support (Engine frame) was built and tested with GShader Library by Akabenko. Thanks to everyone who reports bugs and ideas in the Workshop comments.